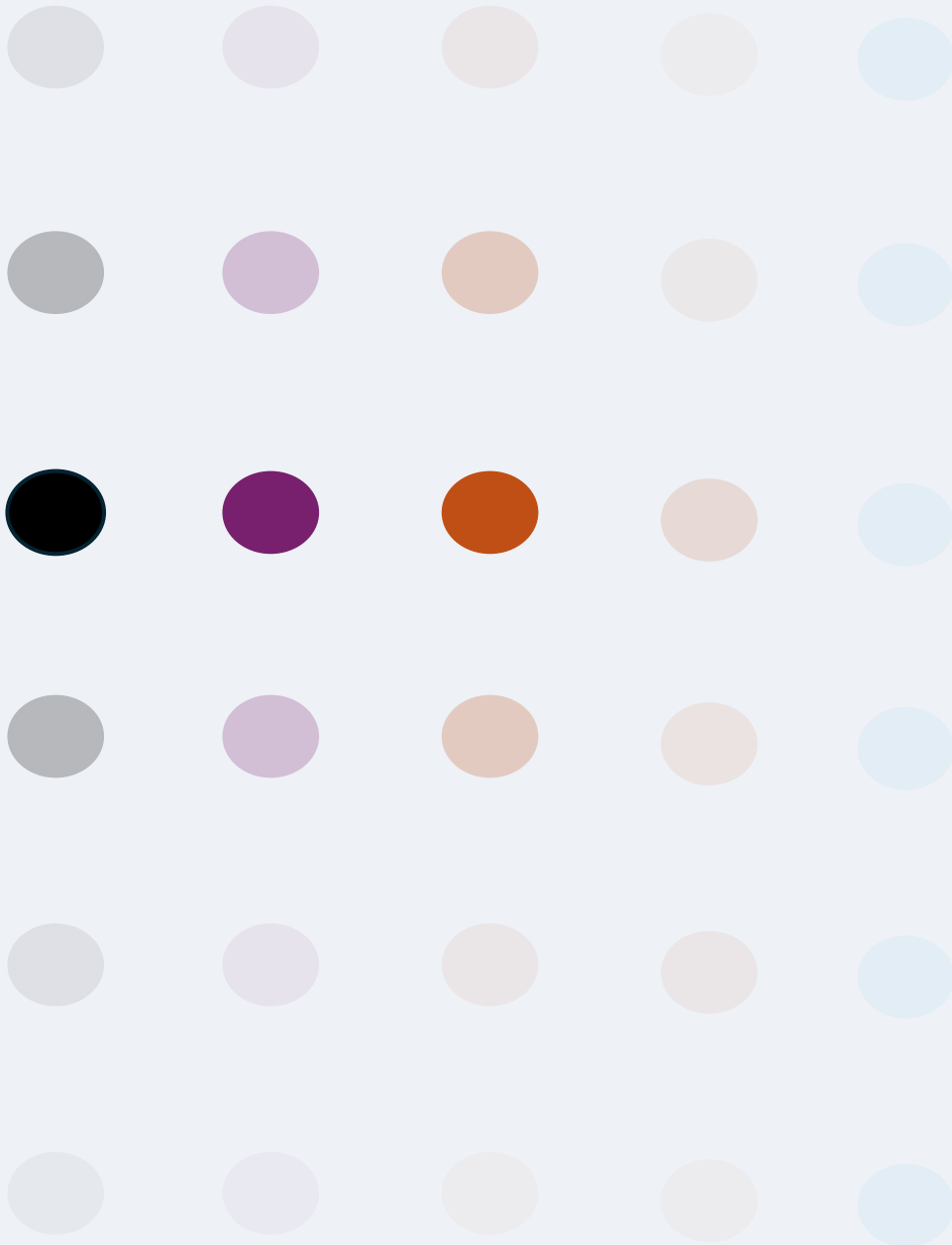


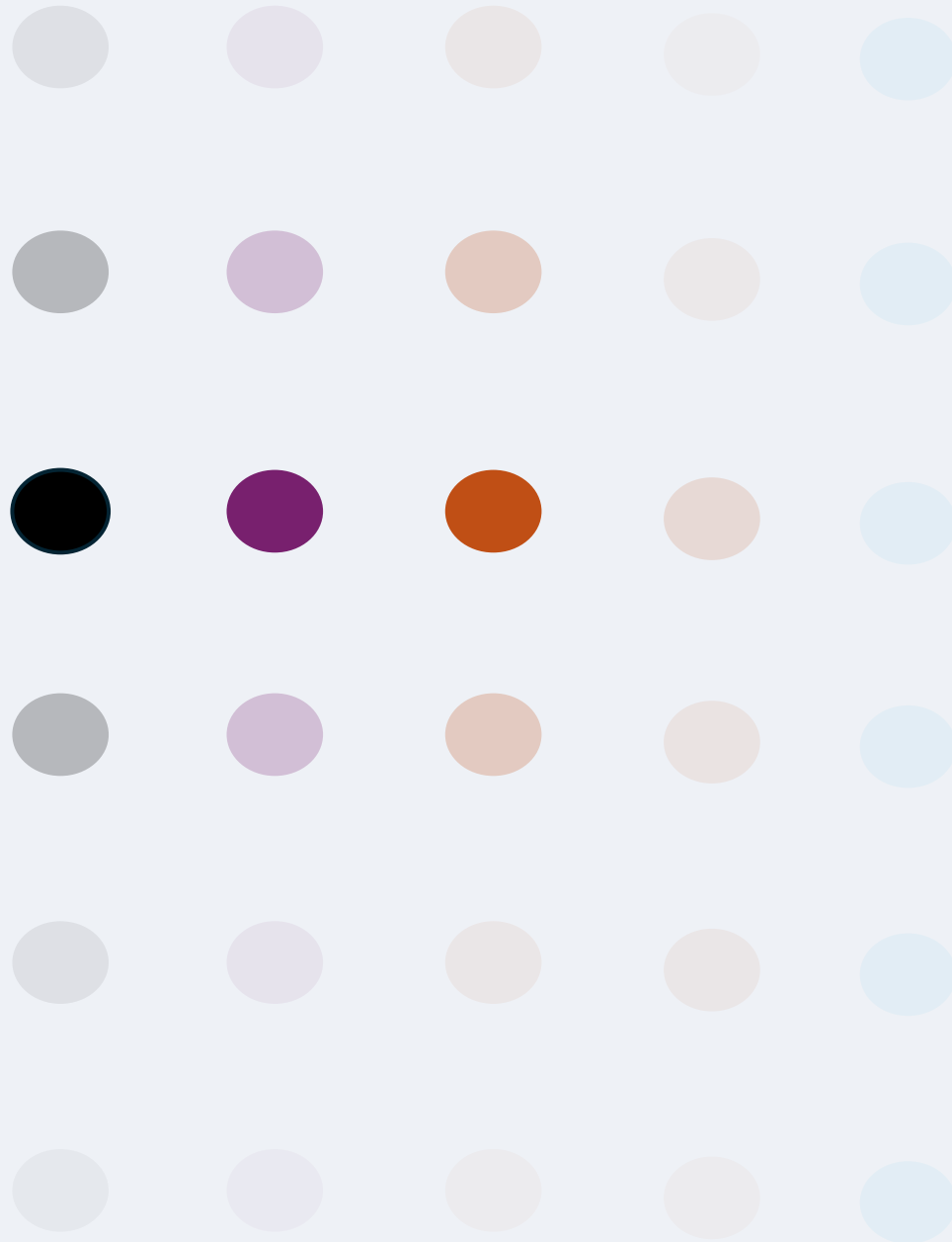


Reticular Local Abstraction (RLA) Conceptual Model, Epistemic CNNs and a PCE bridge

Dott. Gianluca Conte



- 1. Introduction**
- 2. RLA Concept Model**
- 3. Epistemic CNN**
- 4. RLA-ECNN: a bridge to PCE**
- 5. ANNEX**



Introduction

1. INTRODUCTION

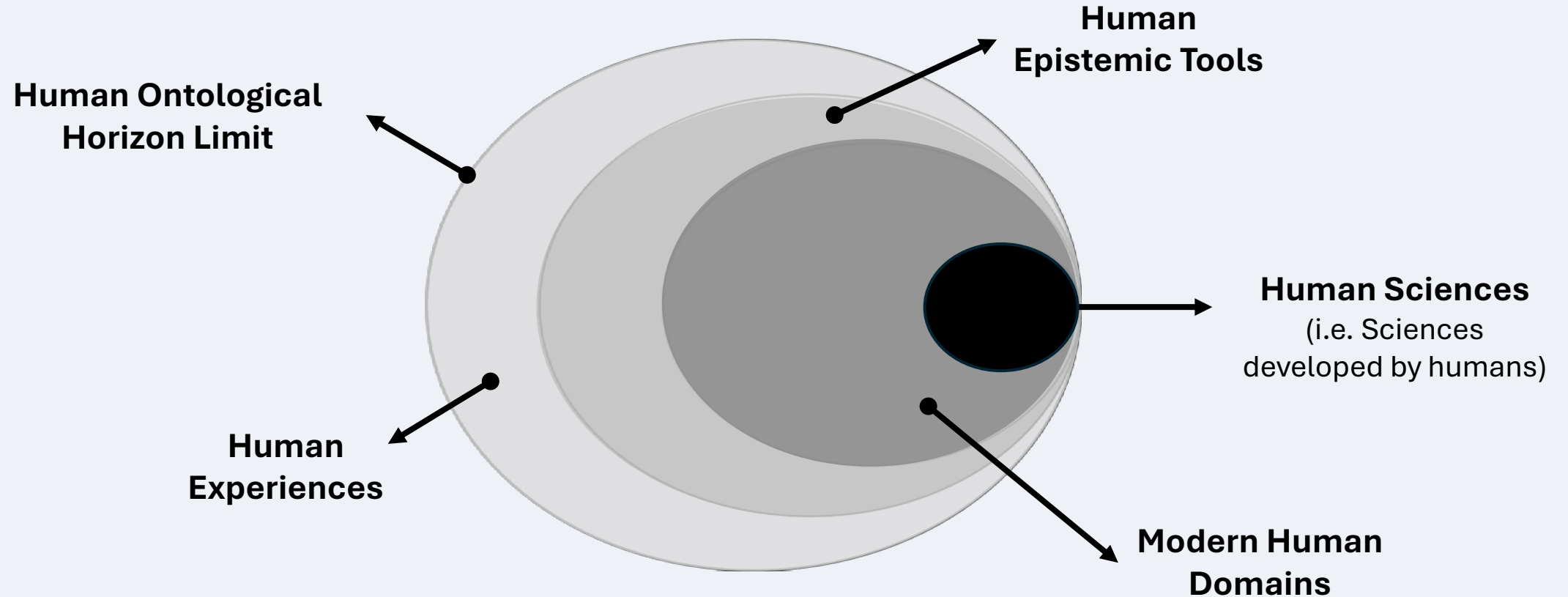
DOMAINS OF HUMAN EXPERIENCE

Domain	Core Idea	Key References
Physical–Natural	Objective order, causality, empirical lawfulness	Kant 1781; Popper 1959; Kuhn 1962; Lakatos 1978
Technical–Technological	Reason applied to creation of artifacts and systems	Descartes 1637; Simon 1969; Floridi 2010; Boon 2018
Social–Human	Rational action, normativity, historical mediation	Hegel 1807; Habermas 1981; Bhaskar 1979; Weimer 2022
Subjective–Phenomenological	Inner experience, intentionality, self-awareness	Descartes 1641; Husserl 1913; Merleau-Ponty 1945; Nagel 1974
Intersubjective–Cultural	Shared meaning, language, symbolic worlds	Wittgenstein 1953; Latour 2005; Barad 2007

A **domain** is a distinct mode of human experience and action — defined by what can be *known*, *done*, and *meant* within its internal logic and limits.

1. INTRODUCTION

DOMAINS OF HUMAN EXPERIENCE



1. INTRODUCTION

HUMAN SCIENCES



Math



Physics



Biology



Chemistry



Economics



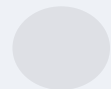
Social Science



Neuroscience



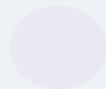
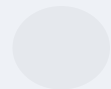
Legal Science



IT



Cognitive Science



1. INTRODUCTION

HUMAN SCIENCES



Physics



Chemistry

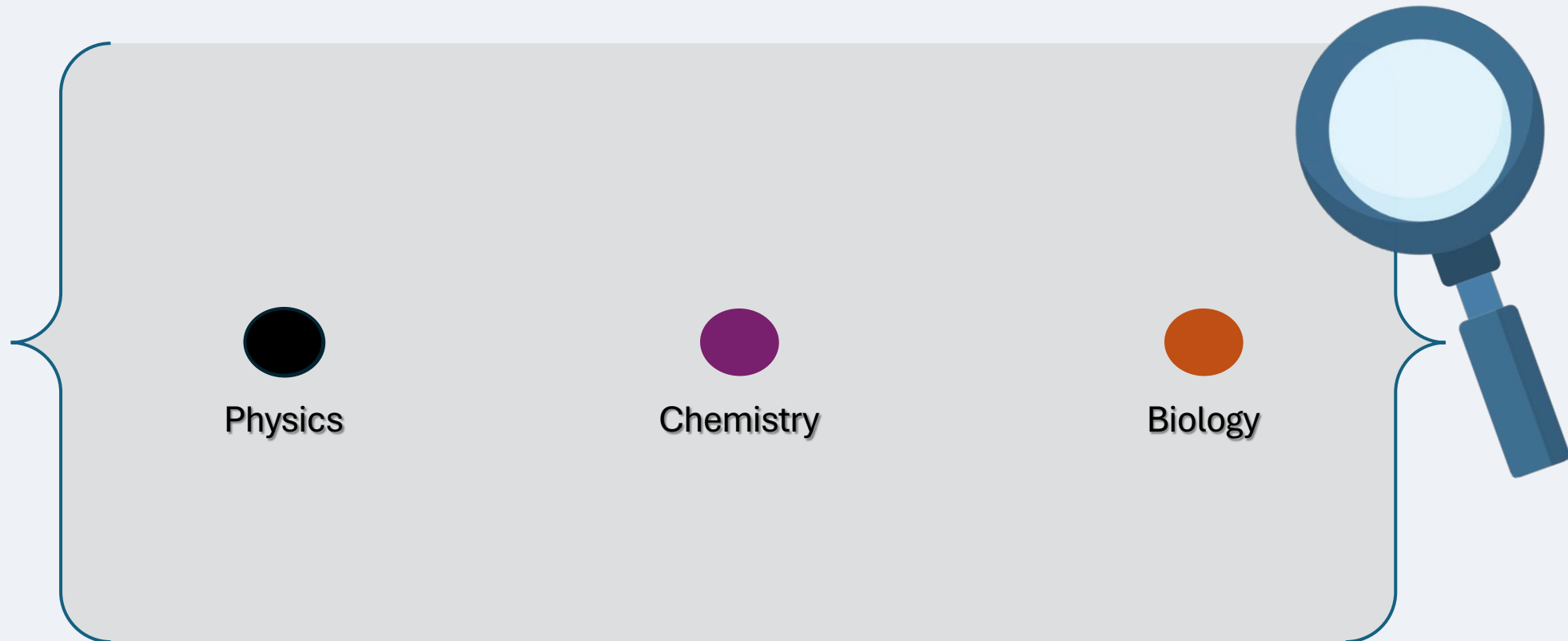


Biology



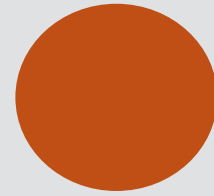
1. INTRODUCTION

HUMAN SCIENCES

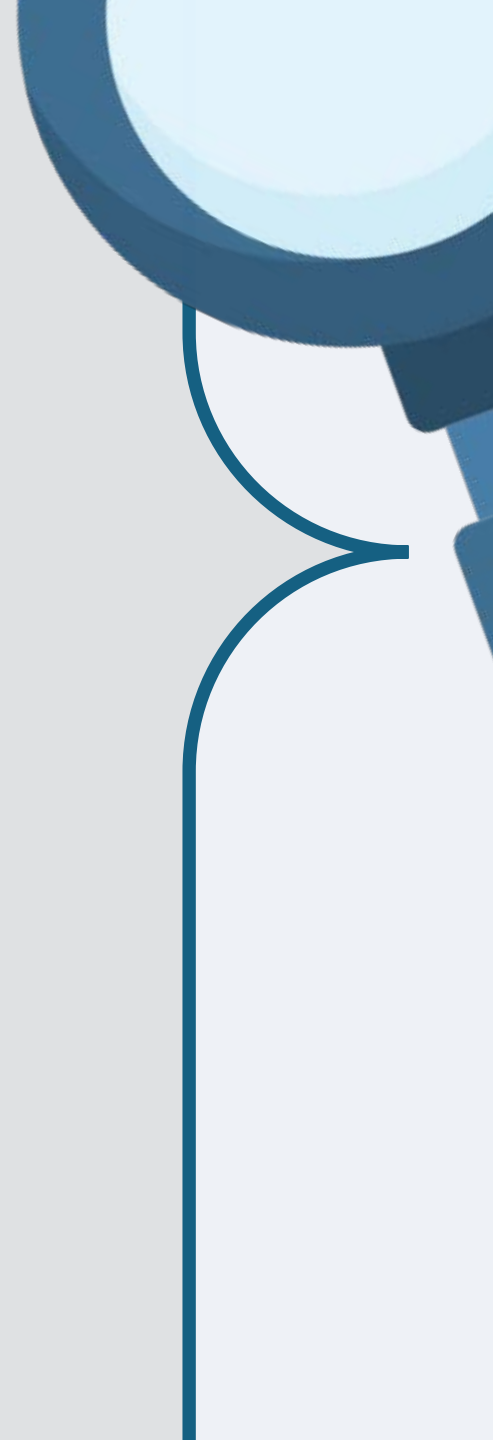


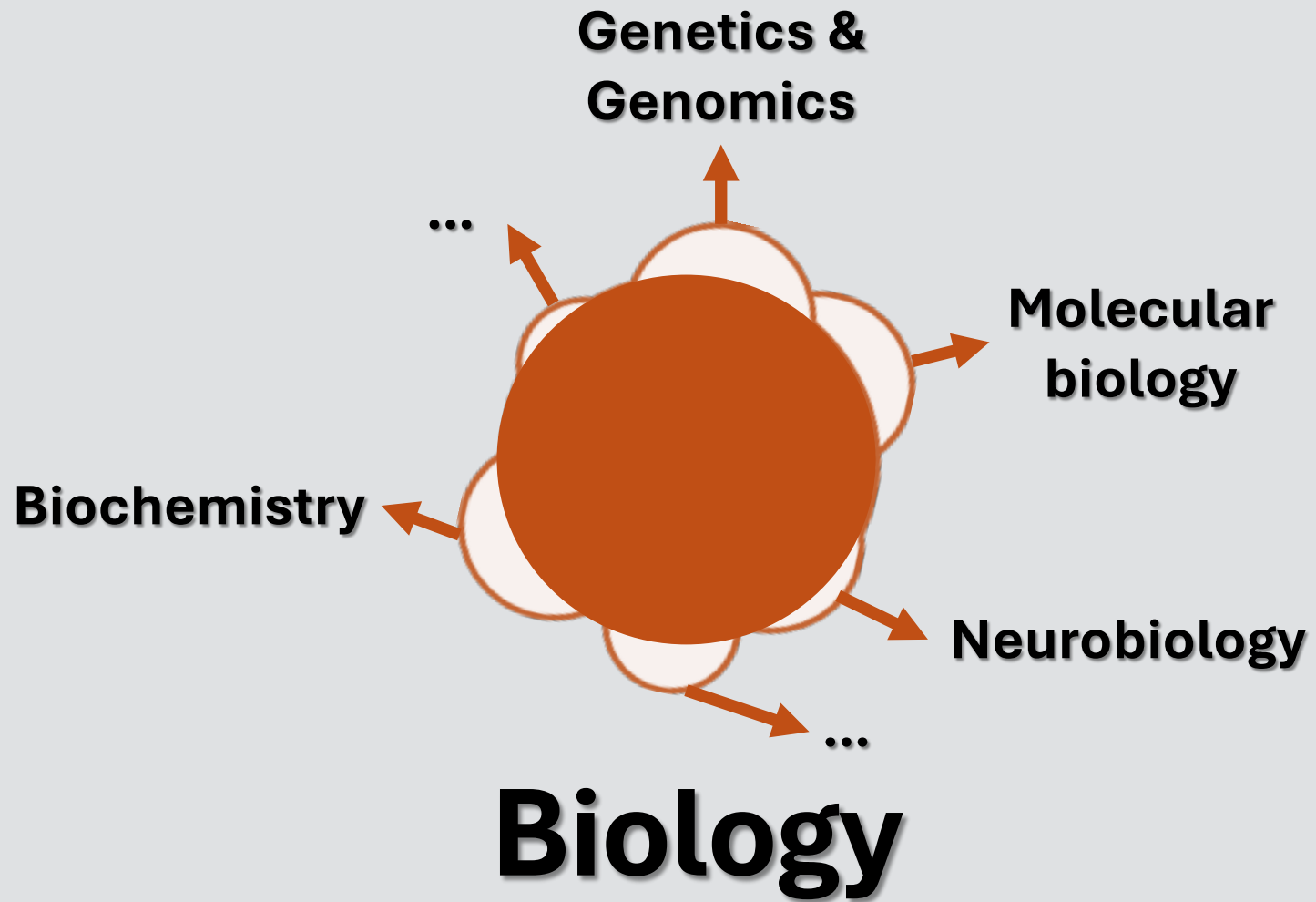


Chemistry



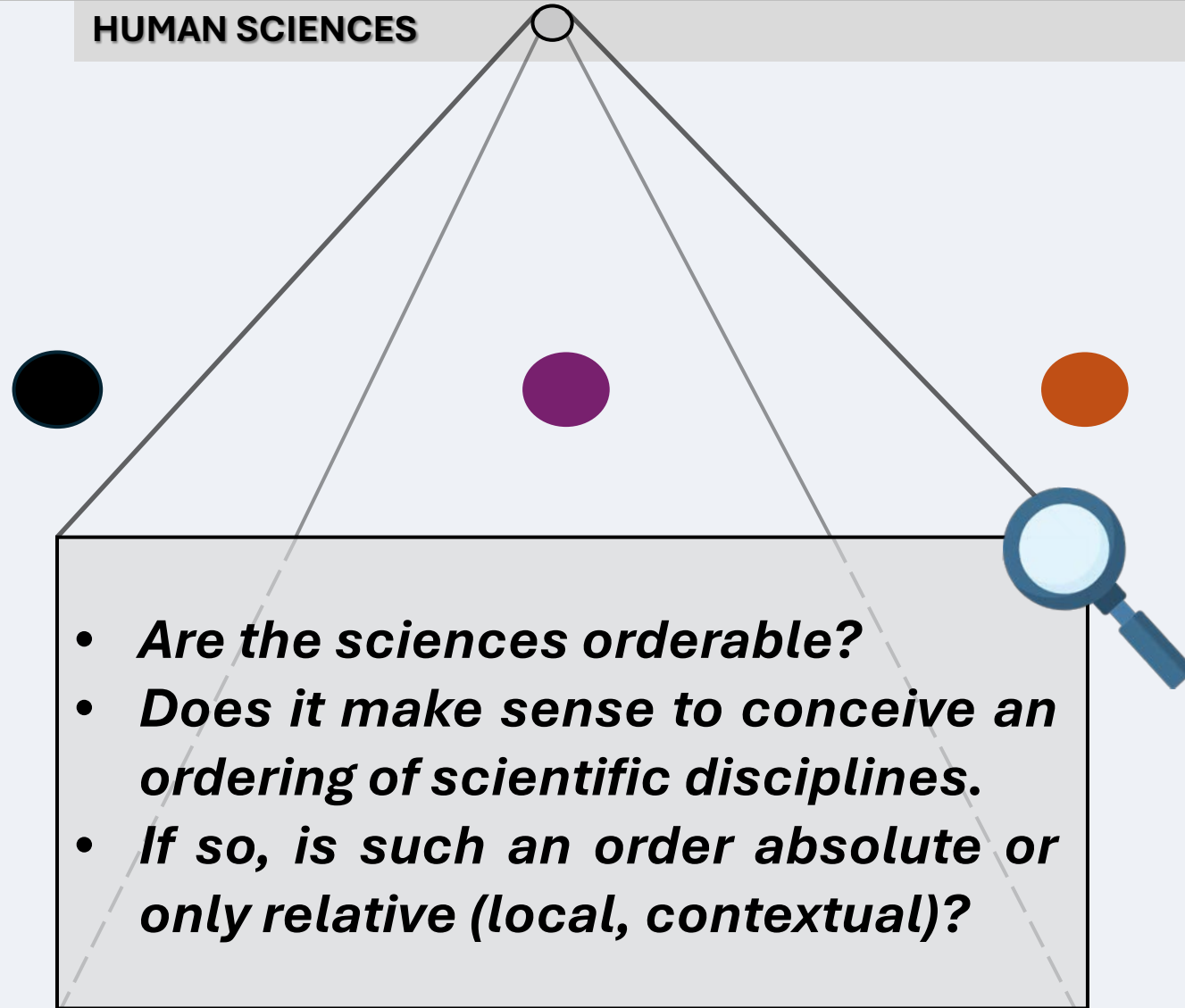
Biology





1. INTRODUCTION

HUMAN SCIENCES

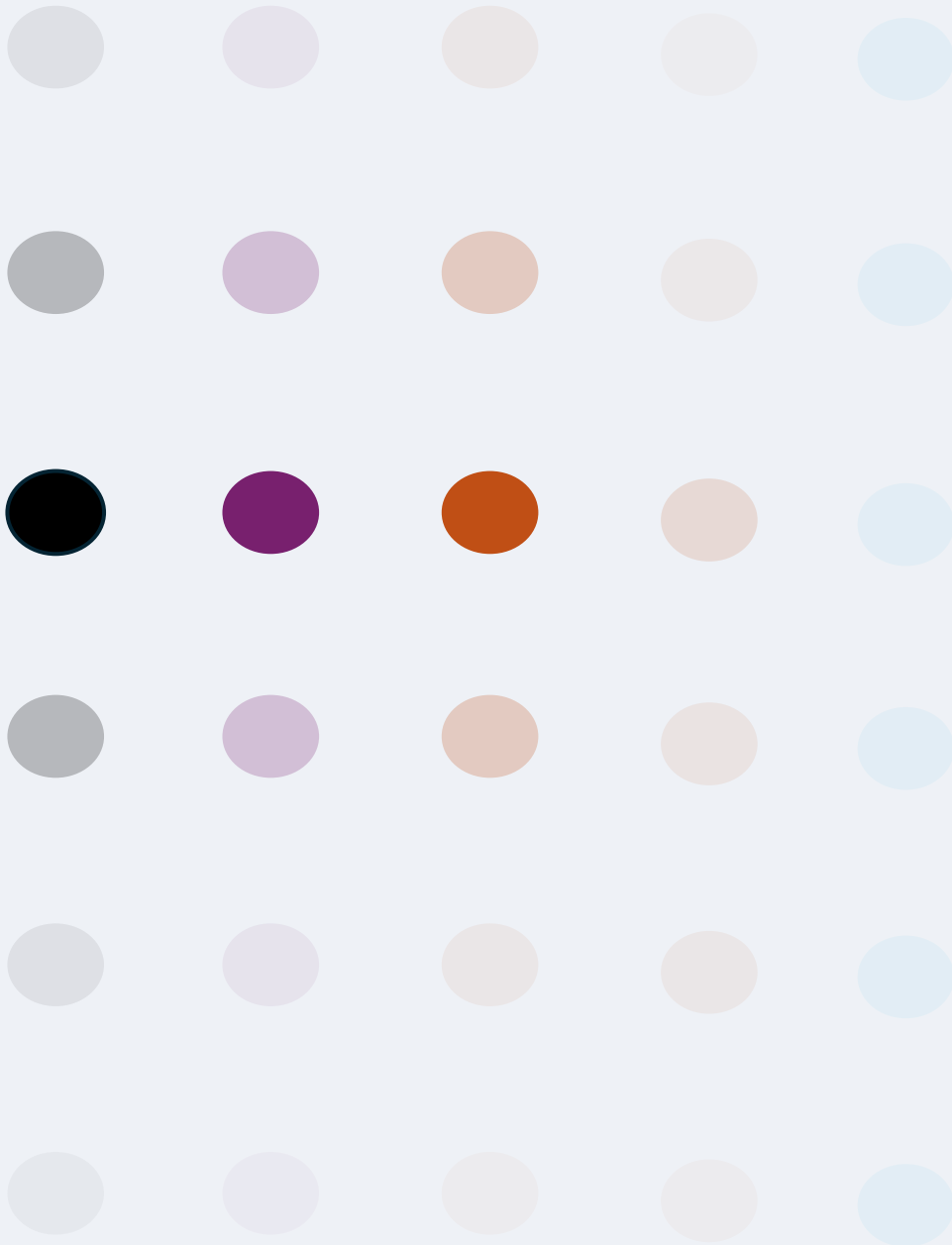


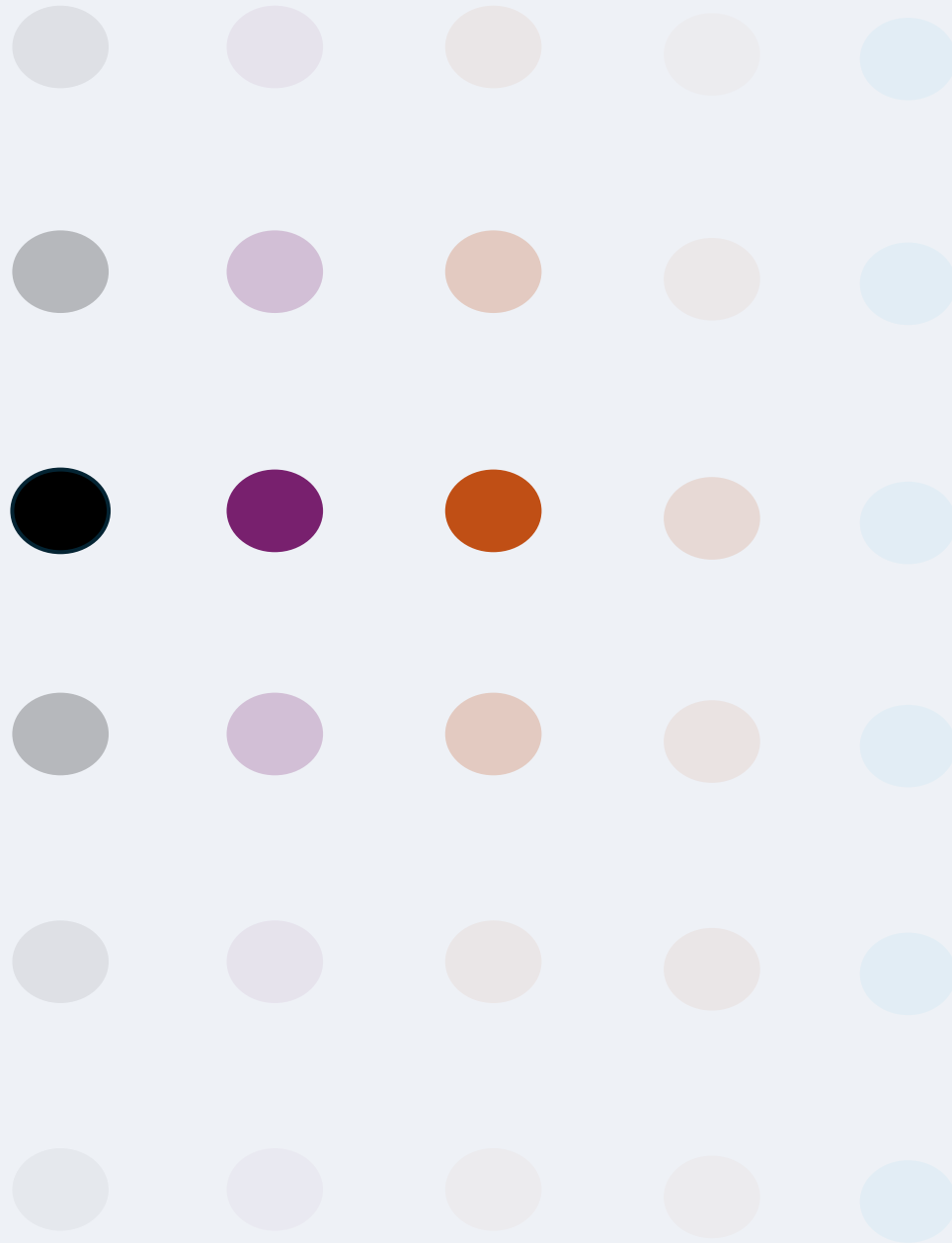
1. INTRODUCTION

HUMAN SCIENCES

View	Epistemic Structure	Key Authors (Date)	Type of Order
Classical	Linear, foundational	Kant (1781), Comte (1830–42)	Absolute
Modern	Contextual, incommensurable	Kuhn (1962), Feyerabend (1975), Lakatos (1978)	Relative / Local
Contemporary	Relational, dynamic	Latour (2005), Barad (2007), Floridi (2010)	Topological / Emergent

The sciences form not a hierarchy, but a reticulum — an evolving topology of domains linked by partial translatability.





RLA Concept Model



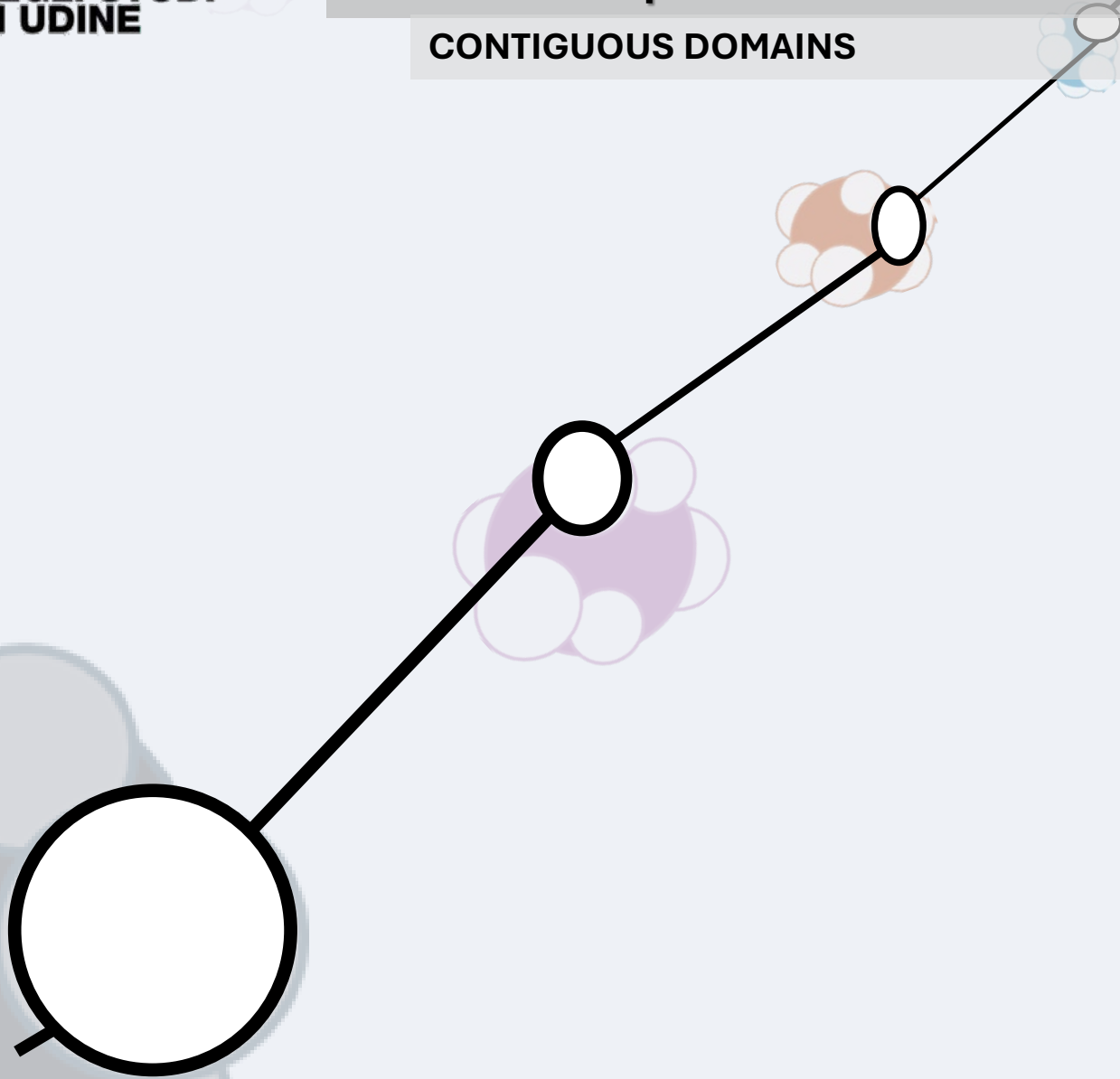
2. RLA Concept Model

CONTIGUOUS DOMAINS



2. RLA Concept Model

CONTIGUOUS DOMAINS



2. RLA Concept Model

CONTIGUOUS DOMAINS

Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. Chicago: University Chicago Press.

Paradigms as local frameworks; partial translatability between disciplines defines epistemic contiguity.

Cartwright, N. (1983). *How the Laws of Physics Lie*. Oxford: Clarendon Press.

Scientific laws are locally valid; bridge laws connect overlapping domains.

Noble, D. (2012). *A theory of biological relativity: No privileged level of causation*. *Interface Focus*, 2(1), 55–64.

No causal hierarchy; biological levels are reciprocally connected — model of epistemic contiguity.

Humphreys, P. (2016). *Emergence: A Philosophical Account*. Oxford: Oxford University Press.

Inter-level dependence formalizes domain adjacency and controlled emergence.

Bechtel, W., & Abrahamsen, A. (2005). *Explanation: A mechanist alternative*. *Studies in History and Philosophy of Biological and Biomedical Sciences*, 36(2), 421–441.

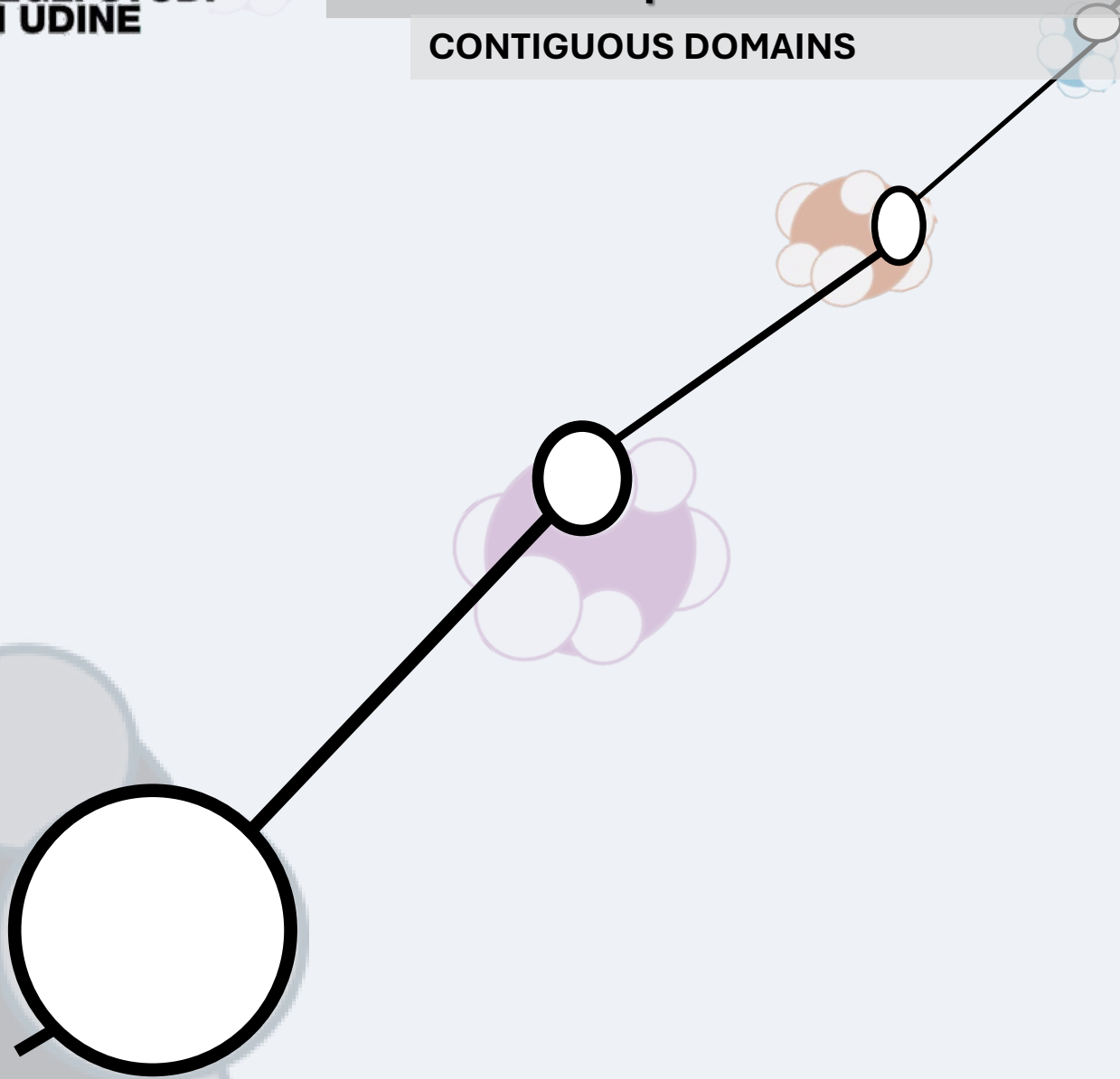
Mechanistic cross-level integration as empirical bridge between models.

Kauffman, S. A. (2000). *Investigations*. Oxford: Oxford University Press.

Reticulate causation and “adjacent possible” describe emergence through non-injective transitions.

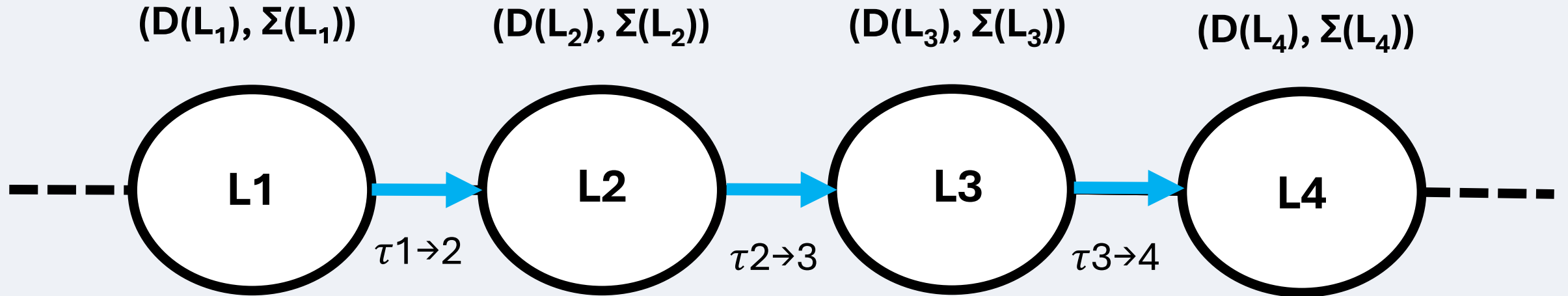
2. RLA Concept Model

CONTIGUOUS DOMAINS



2. RLA Concept Model

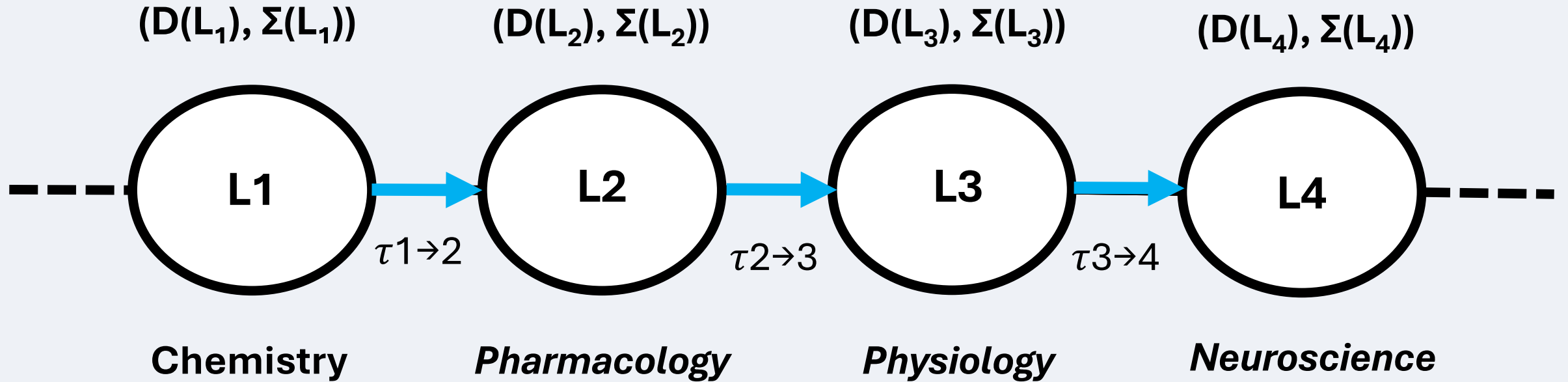
CONTIGUOUS DOMAINS



$$(D(L_n), \Sigma(L_n))$$
$$\tau_{n \rightarrow n+1}$$

2. RLA Concept Model

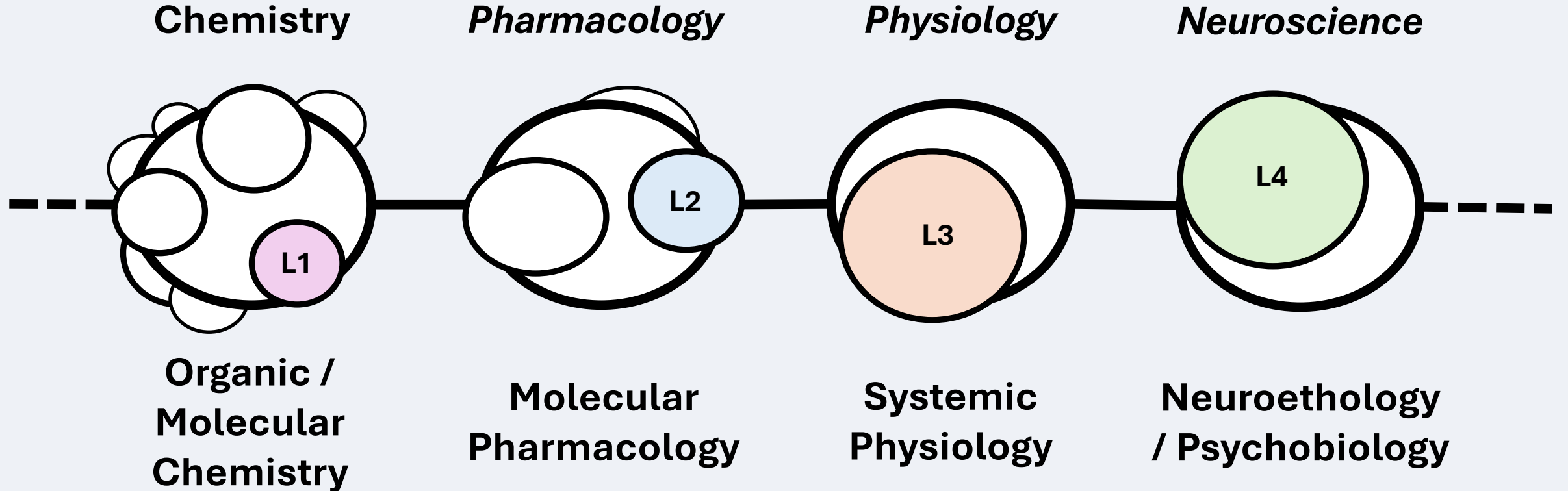
CONTIGUOUS DOMAINS



Conventional Linearization Only

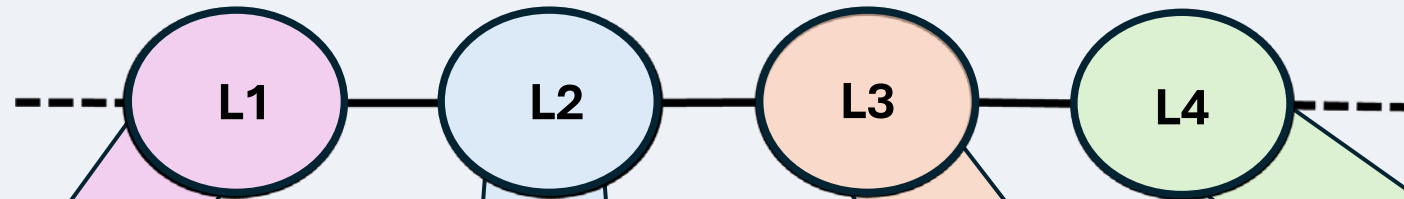
2. RLA Concept Model

CONTIGUOUS DOMAINS



2. RLA Concept Model

CONTIGUOUS DOMAINS



L₁ Organic/Molecular Chemistry

- Domain: Organic & quantum chemistry; structure–function of small molecules.
- Core laws: Quantum mechanics (Schrödinger, DFT), thermodynamics (Gibbs, Arrhenius), chemical kinetics.

Dominant References: Atkins & de Paula, *Physical Chemistry* (OUP, 2023) Leach, *Molecular Modelling* (Pearson, 2013) Cramer, *Essentials of Computational Chemistry* (Wiley, 2013)

L₂ Molecular Pharmacology Domain:

- Drug–target interactions, receptor binding, enzyme inhibition, signaling.
- Core laws: Langmuir binding, Hill equation, Michaelis–Menten kinetics, receptor theory.

Dominant References: Rang & Dale's *Pharmacology* (Elsevier, 2024) Kenakin, *Pharmacology in Drug Discovery* (Academic Press, 2016) Gabrielsson & Weiner, *Pharmacokinetic and Pharmacodynamic Data Analysis* (CRC, 2012)

L₃ Systemic Physiology

- Domain: Integrated organ function, homeostasis, metabolic and hormonal control.
- Core laws: Mass balance ODEs, feedback regulation, compartment and control models.

Dominant References: Guyton & Hall, *Textbook of Medical Physiology* (Elsevier, 2021) Noble, *The Music of Life* (OUP, 2006) Kitano, *Systems Biology* (Science, 2002)

L₄ Neuroethology

- Domain: Neural circuits underlying perception, motivation, and adaptive behavior.
- Core laws: Neural population dynamics (Hopfield, Wilson–Cowan), reinforcement learning, drift–diffusion decision models.

Dominant References: Kandel et al., *Principles of Neural Science* (McGraw-Hill, 2021) Schultz et al., *Science* 275 (1997) — dopamine reward prediction Tinbergen, *The Study Instinct* (OUP, 1951)

Transition

$\tau_{1 \rightarrow 2}$

Type

[...]

Formalism

[...]

Transition

$\tau_{1 \rightarrow 2}$

Type

[...]

Formalism

[...]

Transition

$\tau_{1 \rightarrow 2}$

Type

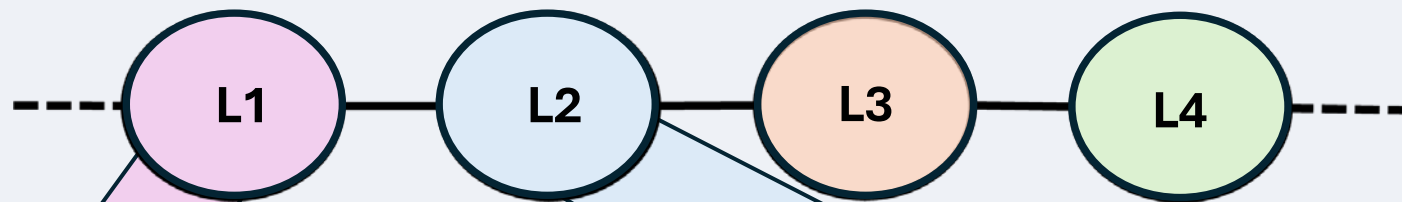
[...]

Formalism

[...]

2. RLA Concept Model

CONTIGUOUS DOMAINS



L₁ Organic/Molecular Chemistry

Domain: Small molecules; functional groups; stereochemistry; 3D conformations; charge distribution; electronic states; physicochemical descriptors (MW, logP, pKa, TPSA, HBD/HBA).

Core Law: Quantum/thermos + cheminformatics:

- Approx. Schrödinger/DFT; Hartree–Fock; semi-empirical MO.
- Reaction kinetics & thermodynamics: Eyring/Arrhenius; Gibbs free energy.
- Conformer generation; tautomer/protomer equilibria.
- Descriptor pipelines & QSAR/QSPR; graph-based featurization.

Transition
 $\tau_{1 \rightarrow 2}$

Type

- Non-Injective
- Occasionally Quasi-injective

Formalism

Docking/MD/ $\Delta\Delta G$; QSAR; free-energy cycles; protonation & permeability models for ADME priors.

L₂ Molecular Pharmacology

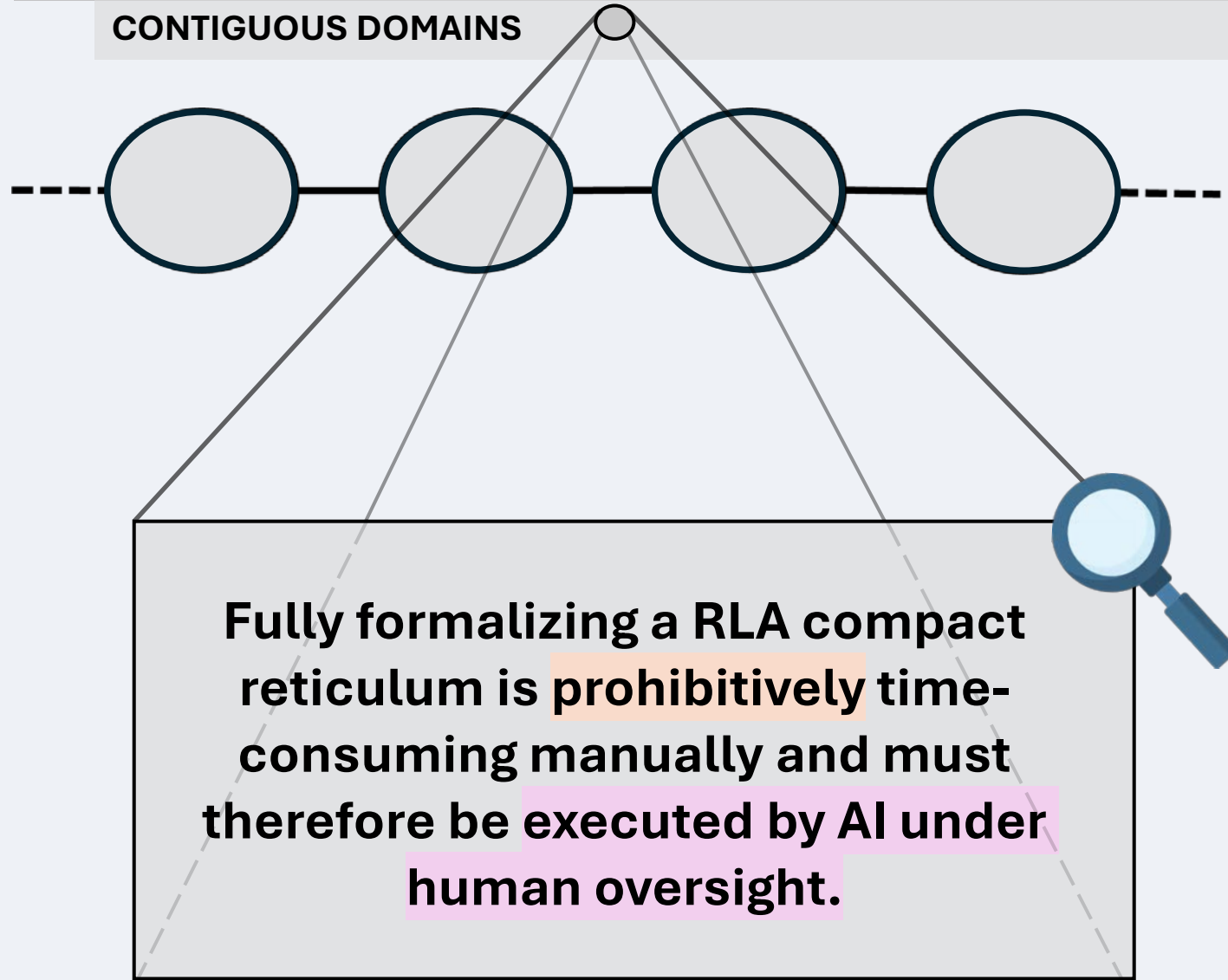
Domain: Targets(receptors/ enzymes/ transporters), ligands, binding sites; affinity/efficacy; occupancy; signaling states; ADME parameters; PK/PD state.

Core Lawes:

- Receptor–ligand binding: mass-action, Langmuir, Hill; $\Delta G \leftrightarrow K_d \Delta G \leftrightarrow K_d$.
- Enzyme kinetics: Michaelis–Menten, competitive/non-competitive inhibition.
- Allostery (Monod–Wyman–Changeux); biased agonism; signal transduction ODEs. PK models (1–3 compartment, clearance/bioavailability) and PD link (E_{max} , sigmoid E_{max} , indirect response).

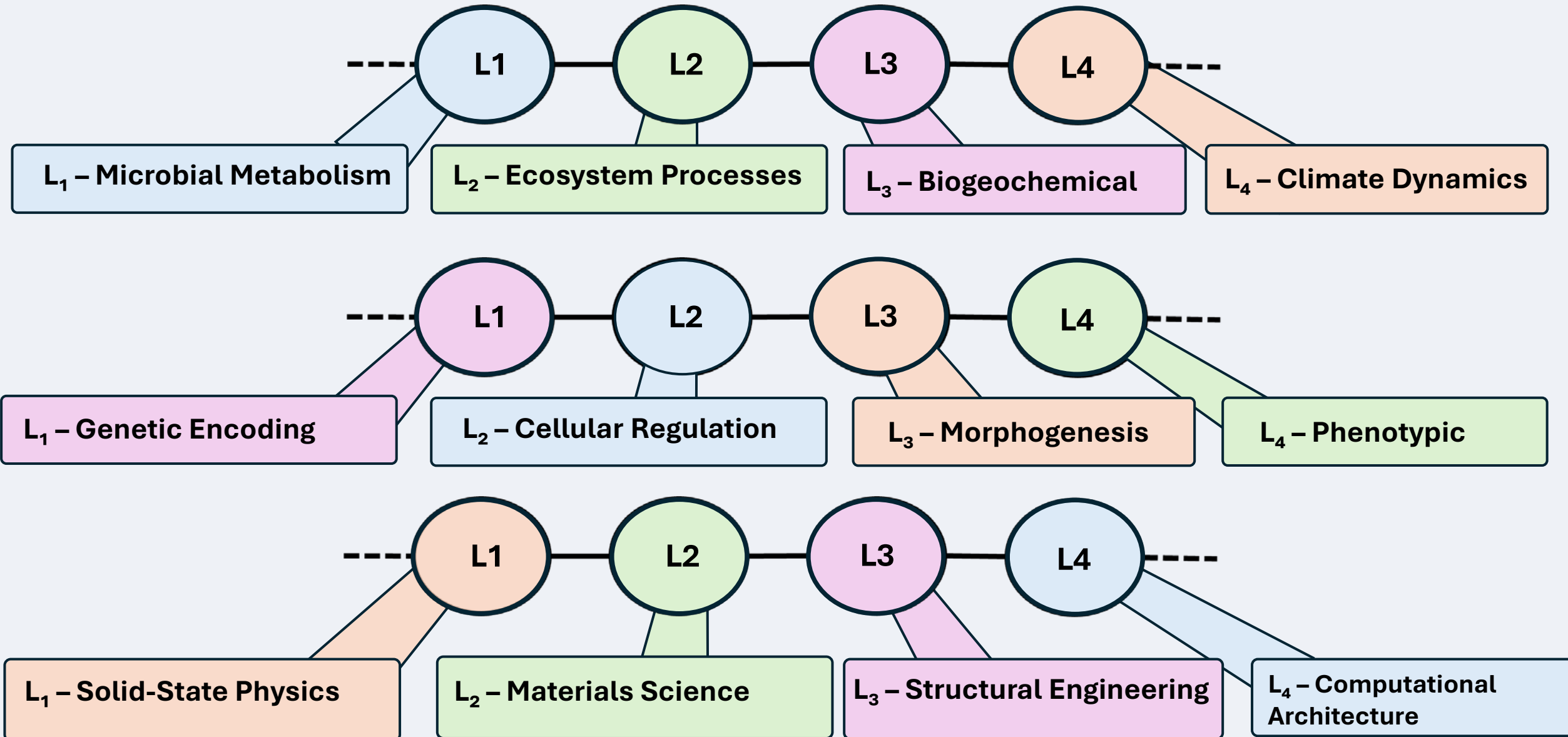
2. RLA Concept Model

CONTIGUOUS DOMAINS



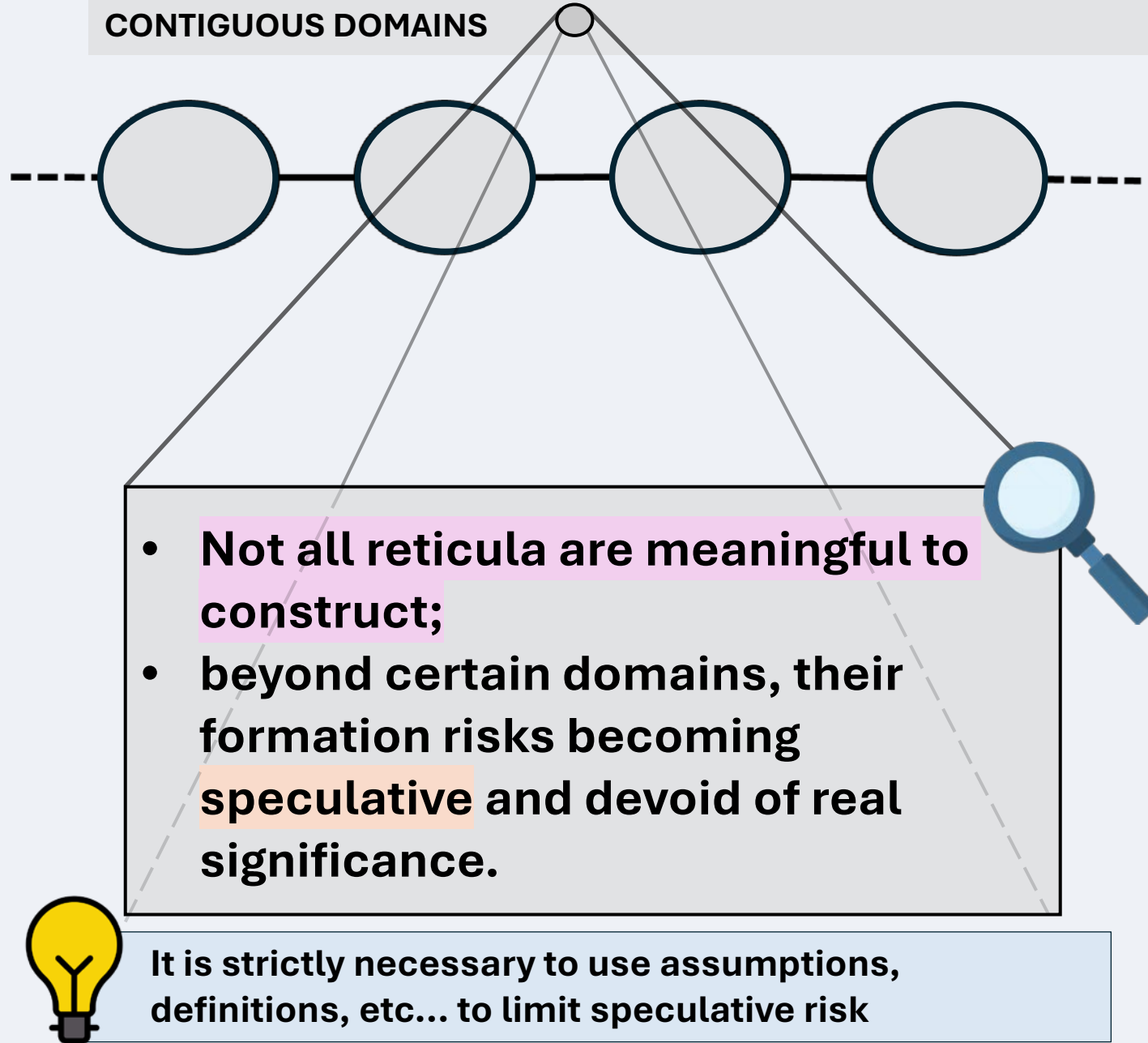
2. RLA Concept Model

CONTIGUOUS DOMAINS (EXAMPLES)



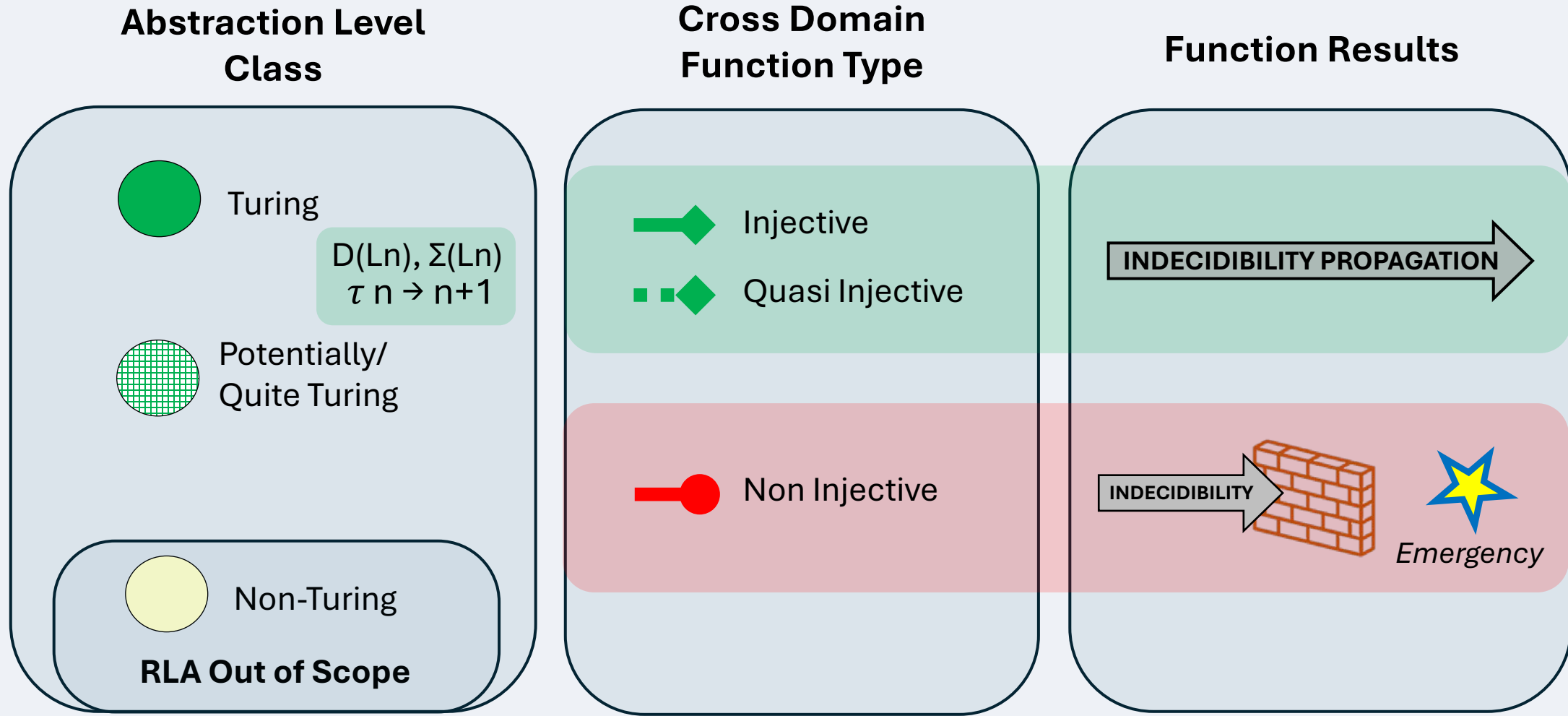
2. RLA Concept Model

CONTIGUOUS DOMAINS



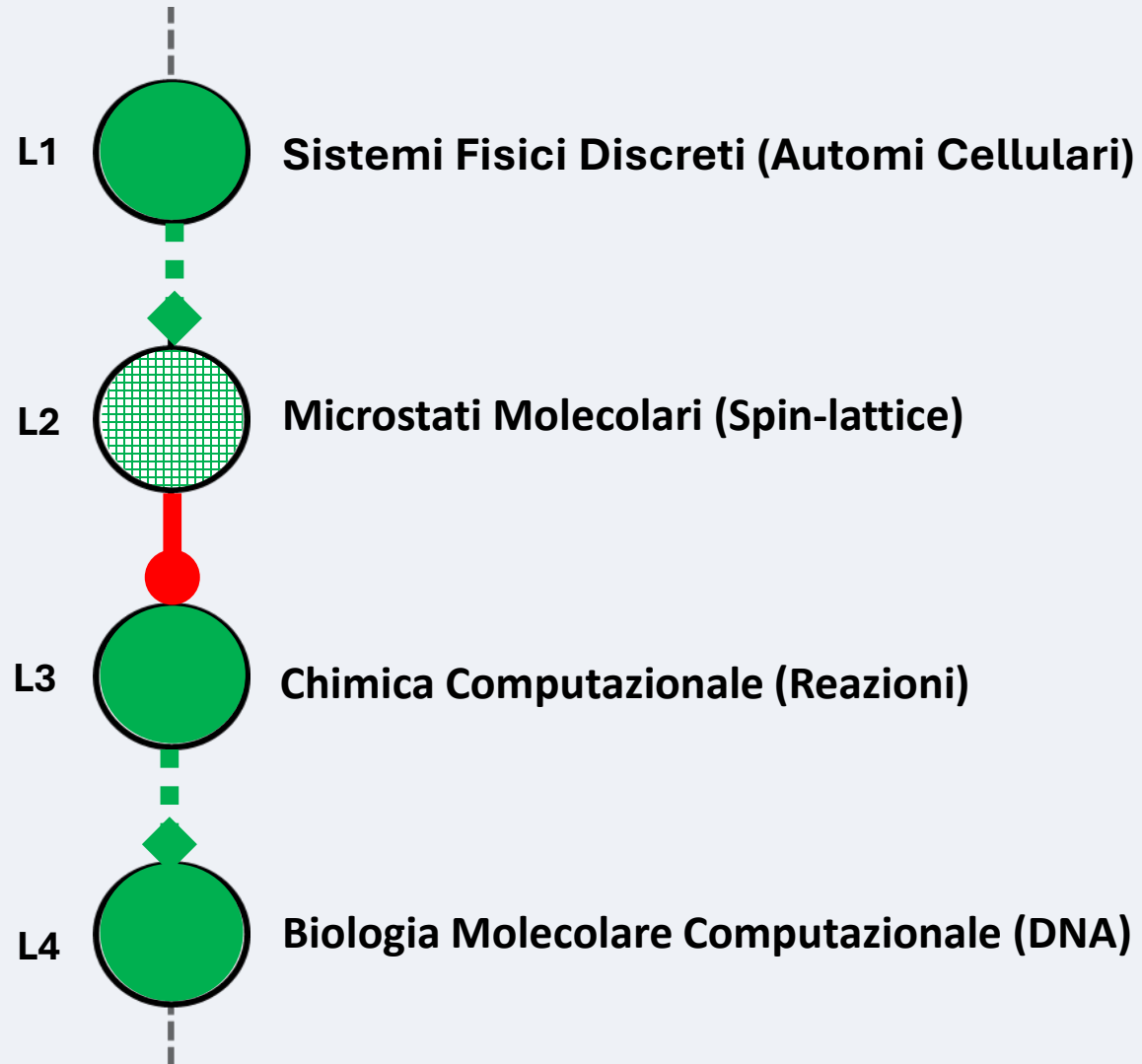
2. RLA Concept Model

DEFINITIONS & ASSUMPTIONS (informal, overview)



2. RLA Concept Model

Biological Being RLA Representation (conceptual example)



Scientific «Supply Chain»

Wolfram 2002

Toffoli & Margolus 1987

Norris 1998

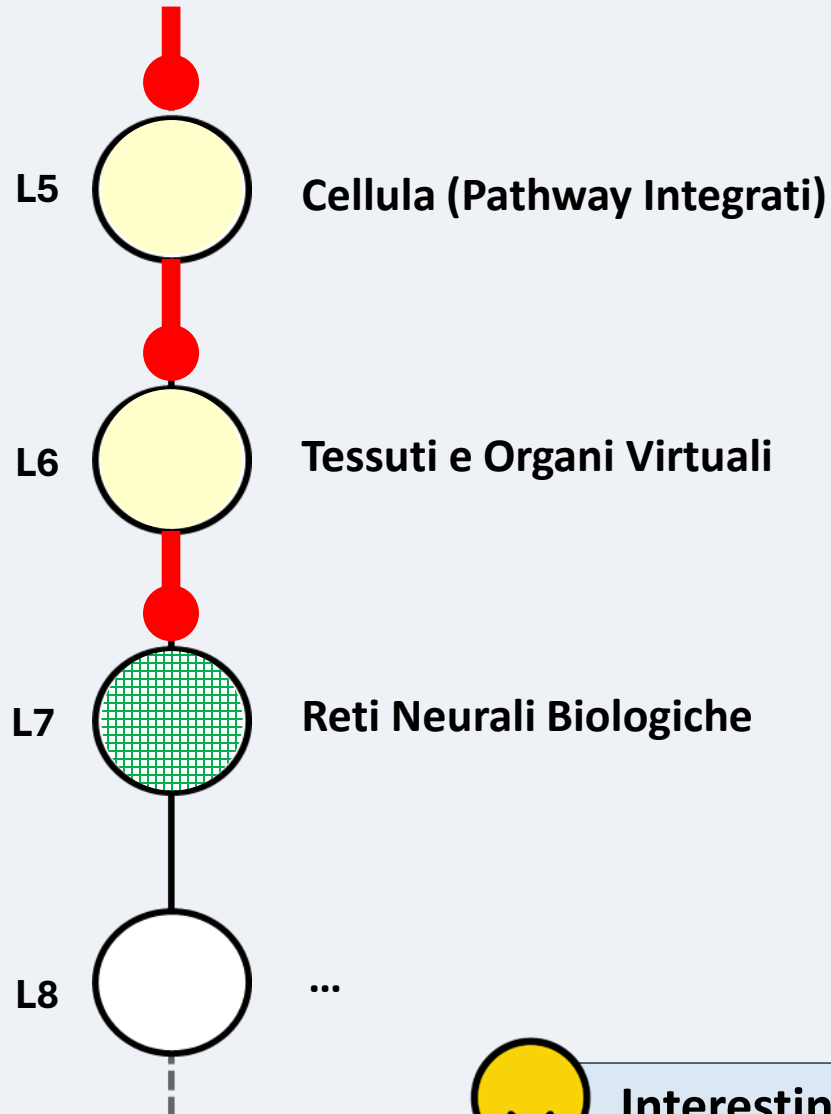
Magnasco 1997

Winfrey 1998

Benenson 2001

2. RLA Concept Model

Biological Being RLA Representation (conceptual example)



**Scientific
«Supply Chain»**

Cardelli 2005

Noble 2008

*Siegelmann &
Sontag 1991*

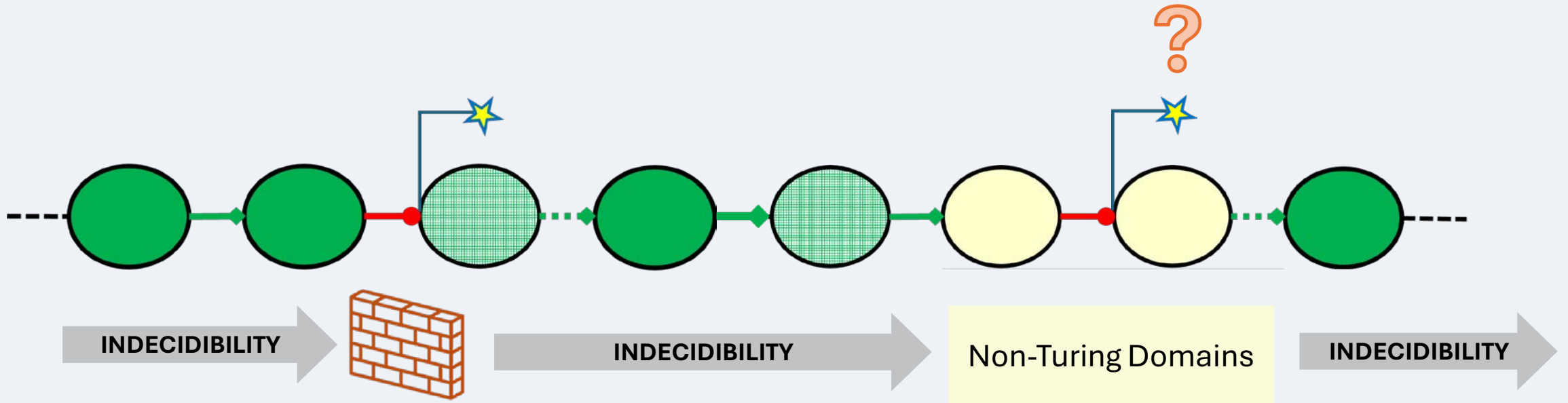
...



Interesting network but not “expressive” enough at every level, so it's just a conceptual exercise

2. RLA Concept Model

DEFINITIONS & ASSUMPTIONS (brief overview)



Formalism is needed

2. RLA Concept Model

DEFINITIONS (informal, overview)



2. RLA Concept Model

FORMAL FRAMEWORK OVERVIEW

Category	Code Range	Count	Conceptual Role
Assumptions (ASS)	ASS-01 → ASS-04	4	Foundational premises ensuring empirical contiguity, Turing-likeness, and domain relevance of collapses
Definitions (DEF)	DEF-01 → DEF-14	14	Formal constructs defining levels, transmissions, computability classes, compactness, and emergence metrics
Axioms (AX)	AX-01 → AX-04	4	Primary laws governing undecidability, propagation, non-injective emergence, and epistemic closure
Lemmas (LEM)	LEM-01 → LEM-03	3	Auxiliary results on composition, reactivation, and quotient interpretation of state pooling
Theorems (THEO)	THEO-01 → THEO-04	4	Demonstrated consequences on propagation barriers, definability, irreversibility, and upper-level undecidability
Corollaries (COR)	COR-01 → COR-02	2	Derived implications on multi-level undecidability propagation and cumulative emergence

2. RLA Concept Model

DEFINITIONS (informal, overview)

DEF-01 — Level of abstraction. $L = \langle D, (L)\Sigma(L) \rangle$ with computable rules acting on a state space.

DEF-02 — Contiguity & transmission. $\tau_{i \rightarrow i+1}: D(L_i) \rightarrow \mathcal{P}(D(L_{i+1}))$ may be non-injective and is discipline-recognized.

DEF-03 — Turing-likeness & C_L . A level is Turing-like if it can simulate a UTM; C_L marks the Turing-critical subset.

DEF-04 — (Non)injectivity classes. Injective / quasi-injective on a subset / non-injective (collapse) are distinguished.

[...]

DEF-09 — Observation predicates & definability. A macro-property at L_{i+1} is definable from L_i if there exists a predicate on L_i matching all preimages; else it is emergent.

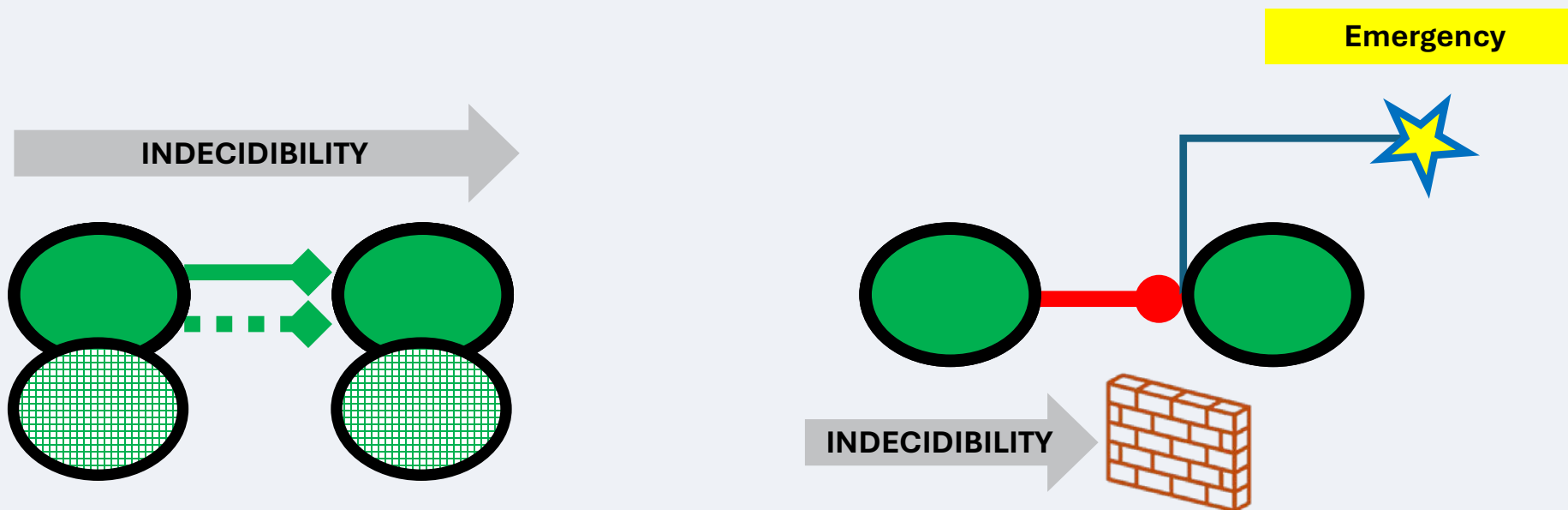
DEF-10 — Relevance function. A domain-specific score s and threshold σ determine when a collapse is “relevant.”

[...]

DEF-14 — Reticular path, loop, morphism. Paths/loops across levels; morphisms preserve levels and transmissions up to relabelling.

2. RLA Concept Model

DEFINITIONS (informal, overview)



2. RLA Concept Model

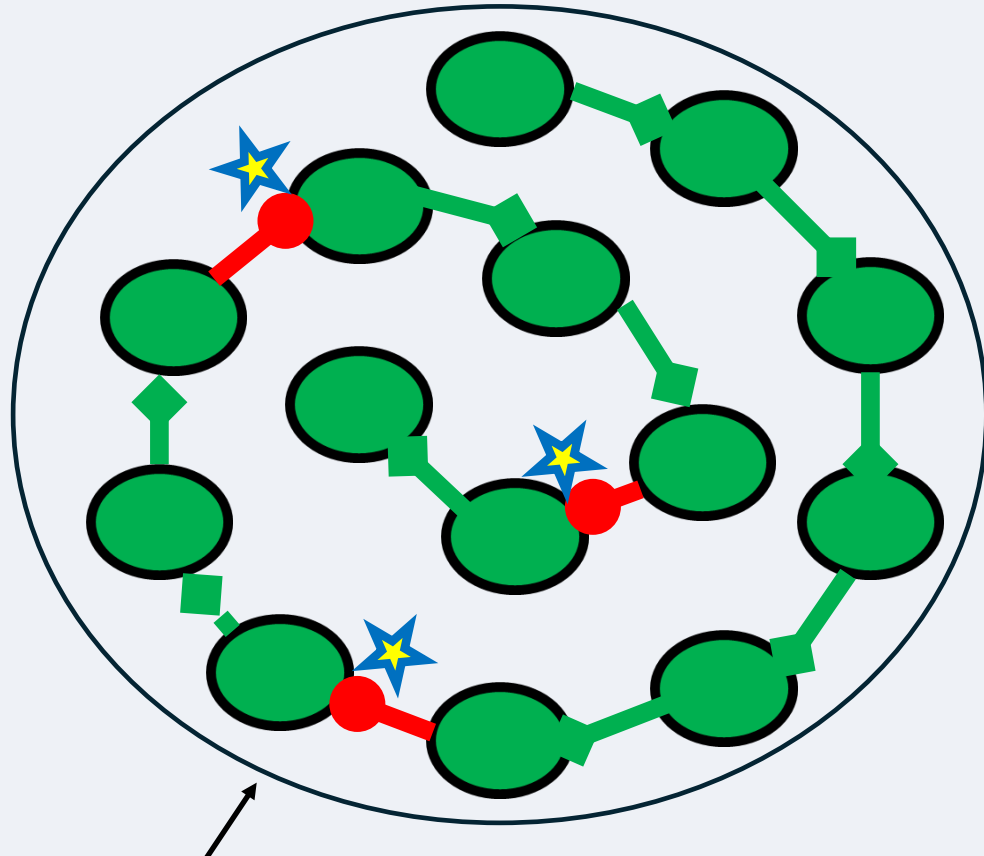
Compat RLA Engineering

Phase	Action	Output
Scouting	<i>Systematic literature review & source curation</i>	<i>Reviewed bibliographic corpus</i>
Extraction	<i>Parse rules, variables, causal links</i>	<i>Structured knowledge base</i>
Abstraction Build	<i>Generate layered $\langle L_1 \dots L_n \rangle + \tau$-mappings</i>	<i>Draft reticular hierarchy</i>
E2E Computability	<i>Every layers and every τ must be turing</i>	<i>Verified compact topology</i>
Metrics	<i>Compute CC, EI, RIR</i>	<i>Quantified model metrics</i>

A bibliographically anchored, **AI-constructed** and **epistemically validated** compact RLA topology—ready for empirical or computational deployment.

2. RLA Concept Model

Compat RLA Engineering



External Collapsed
Representation Horizon

- If a domain is **Turing-compatible** (DEF-07) and adjacent domains are linked by computable, empirically grounded **transmissions** (DEF-02, ASS-02), the chain forms a **compact reticulum globally computable** (DEF-05), (DEF-06), (DEF-07/08), with a **non-injective boundary** filtering external inputs (AX-04) (no external oracle, no epistemic perturbation).
- Synthetic observations within the reticular horizon are expected to be **epistemically indistinguishable** from expert human observations — a direct consequence of **undecidability propagation** (AX-01/02) and **emergence from collapse** (AX-03, THEO-02).

2. RLA Concept Model

Compat RLA Engineering

[...]

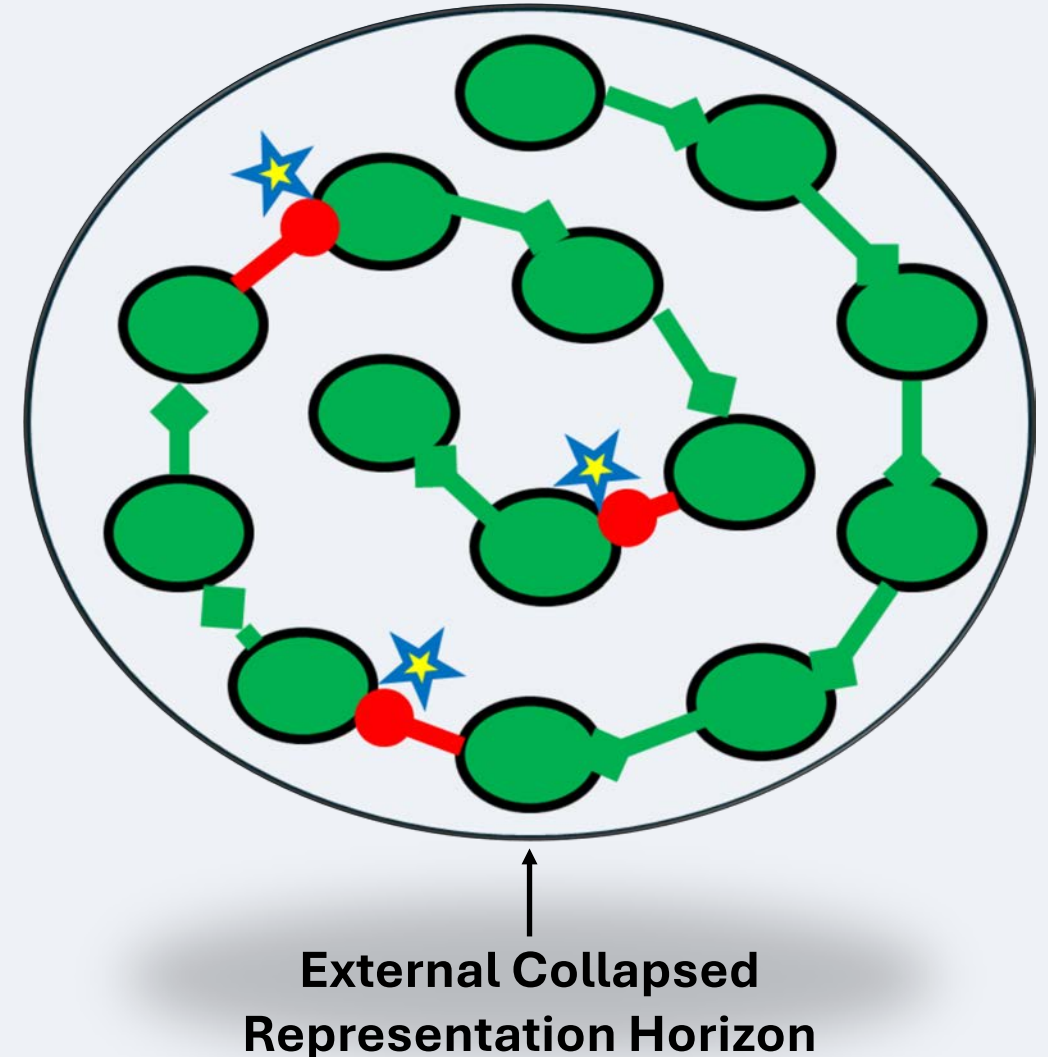
DEF-05 — Ontological Independence (OI). Explanatory entities/laws are internal; external inlets are filtered non-injectively.

DEF-06 — Epistemic Closure (EC). Internal phenomena are explainable from internal levels and transmissions.

DEF-07 — Turing-Computability (TC). All rules/transmissions are algorithmically implementable and simulable.

DEF-08 — Compact RLA. OI, EC, and TC jointly hold.

[...]



2. RLA Concept Model

Compat RLA Engineering



**Generalistic Bryophyta
(mosses)**

Category	Quantitative Summary
Total references	156
Temporal coverage	≈ 50 years (1973–2023)
Type distribution	
– <i>Peer-reviewed journal articles</i>	118 (≈75%)
– <i>Academic books / book chapters</i>	16 (≈10%)
– <i>Technical / institutional reports</i>	14 (≈9%)
– <i>Preprints / online resources</i>	8 (≈5%)
Disciplinary spread	
– <i>Plant physiology & bryology</i>	≈35%
– <i>Molecular & epigenetic biology</i>	≈20%
– <i>Ecology & hydrology</i>	≈15%
– <i>Systems biology & modeling</i>	≈15%
– <i>Philosophy & epistemology</i>	≈10%
– <i>Climate / geophysical datasets</i>	≈5%

2. RLA Concept Model

Compat RLA Engineering

[L20] ↔ Meta-Systemic (Health/Resilience)

↑ ↓

[L19] Evolutionary Plasticity
[L18] Tissue Mechanics
[L17] Geometric Patterning
[L16] Intercellular Communication
[L15] Senescence / Turnover
[L14] Biological Rhythms
[L13] Soil / Substrate Exchange
[L12] Systemic Control (Homeostasis)
[L11] Epigenetic Memory

...

→ ...

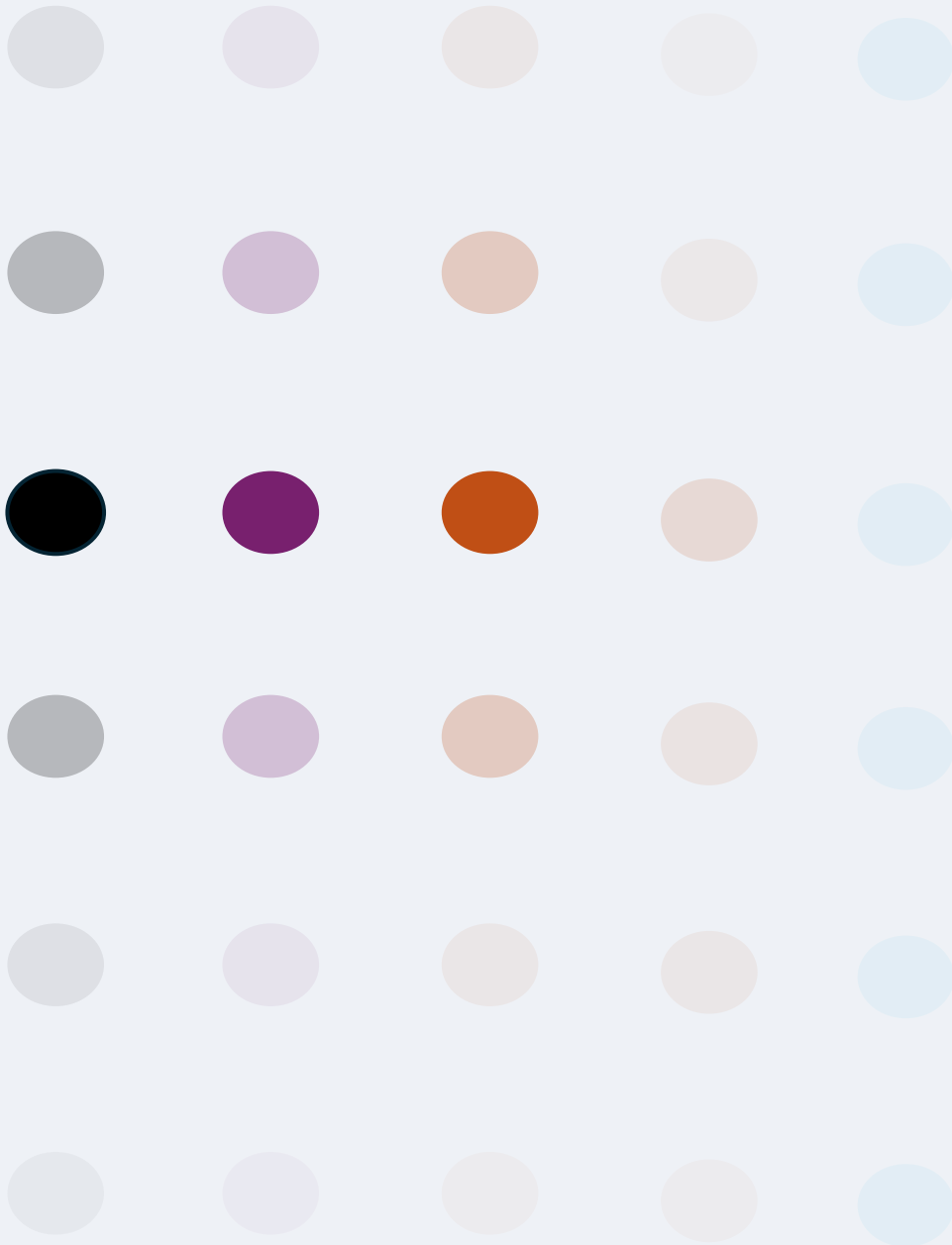
[L10] Macroclimate Context
[L09] Local Environment
[L08] Biotic Interactions
[L07] Reproduction Cycle
[L06] Morphogenesis
[L05] Tissue Architecture
[L04] Cell Physiology
[L03] Metabolic Regulation
[L02] Transcription / Translation
[L01] Genomic–Epigenetic Core

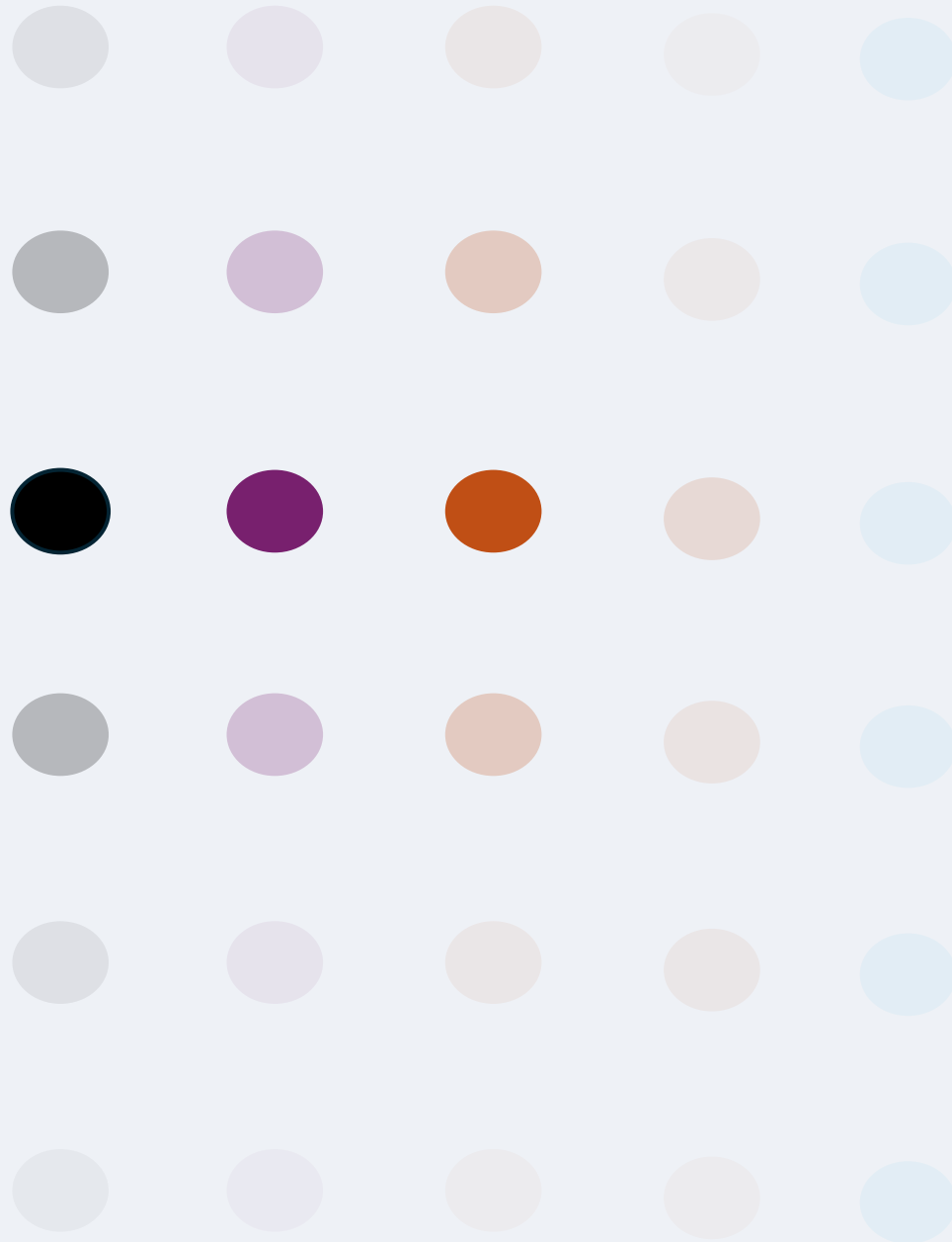
↓ ↑

Bottom-up emergence Top-down regulation

42 biophysical parameters · 38 dynamic equations · 10 causal relations · 30 feedback actions ·

22 collapse–emergence metrics · 2 composite indices · 4 climatic drivers (T°, light, rainfall, trend)





RLA Final Notes

2. RLA Concept Model

Final Notes

Abstraction makes the world intelligible (without a single ladder)

RLA key points: DEF-01 (level as domain + rules), DEF-02 (adjacent relations), ASS-02 (empirical contiguity), DEF-03 (Turing vs non-Turing levels).

Three ways information travels: injective, quasi-injective, collapsing

RLA key points: DEF-04 (injective / quasi-injective / non-injective), ASS-04 (relevant collapse).

Emergence: novelty is lawful and measurable

RLA key points: DEF-09 (definability), DEF-11–13 (CC, IE, RIR), AX-03 (emergence from collapse), THEO-02–03 (non-derivability; no total left-inverse), COR-02 (accumulated emergence).

A web of levels: how domains talk to each other

RLA key points: DEF-02 (transmissions across levels), DEF-14 (paths/loops/morphisms), ASS-02 (justified contiguity).

Computation sets the frontier; undecidability can propagate

RLA key points: ASS-03 (Turing-critical subset), AX-01 (undecidability in Turing-like levels), AX-02 (propagation via quasi-injectivity), LEM-01 (composition), LEM-02 (re-activation).

Compact systems: coherence you can test, not totality

RLA key points: DEF-05 (Ontological Independence), DEF-06 (Epistemic Closure), DEF-07 (Computability), DEF-08 (compactness), AX-04 (closure).

2. RLA Concept Model

Final Notes

Knowledge is structurally finite, yet endlessly extensible

***Inference:** From ASS-03, DEF-03, AX-01, COR-01 — undecidability sets intrinsic limits; reticular connectivity allows unbounded recombination.*

Emergence is not mystery but architecture

***Inference:** From DEF-04, AX-03, THEO-02–03, COR-02 — relevant non-injectivity produces novel properties; emergence is structurally necessary.*

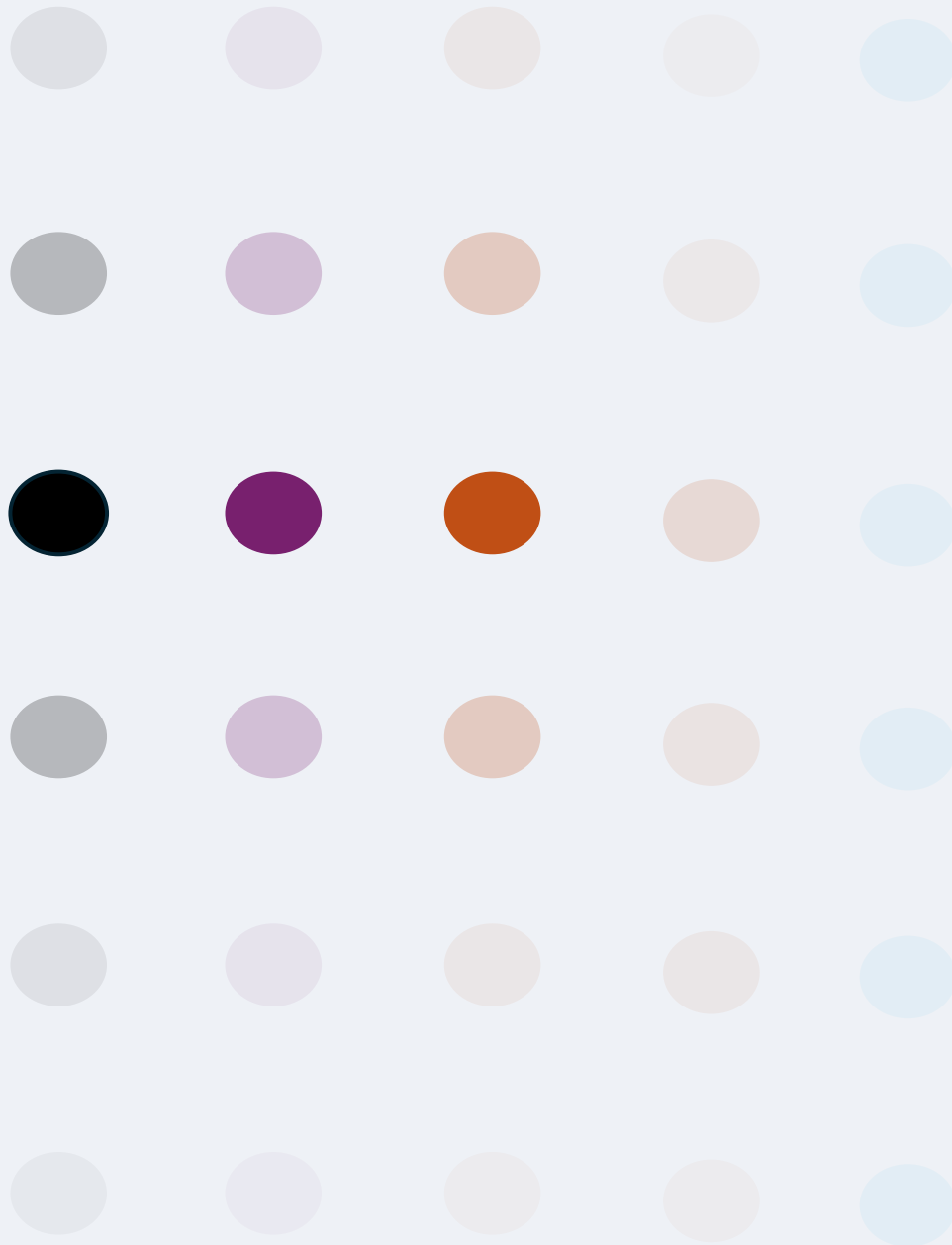
Truth is relational, not absolute

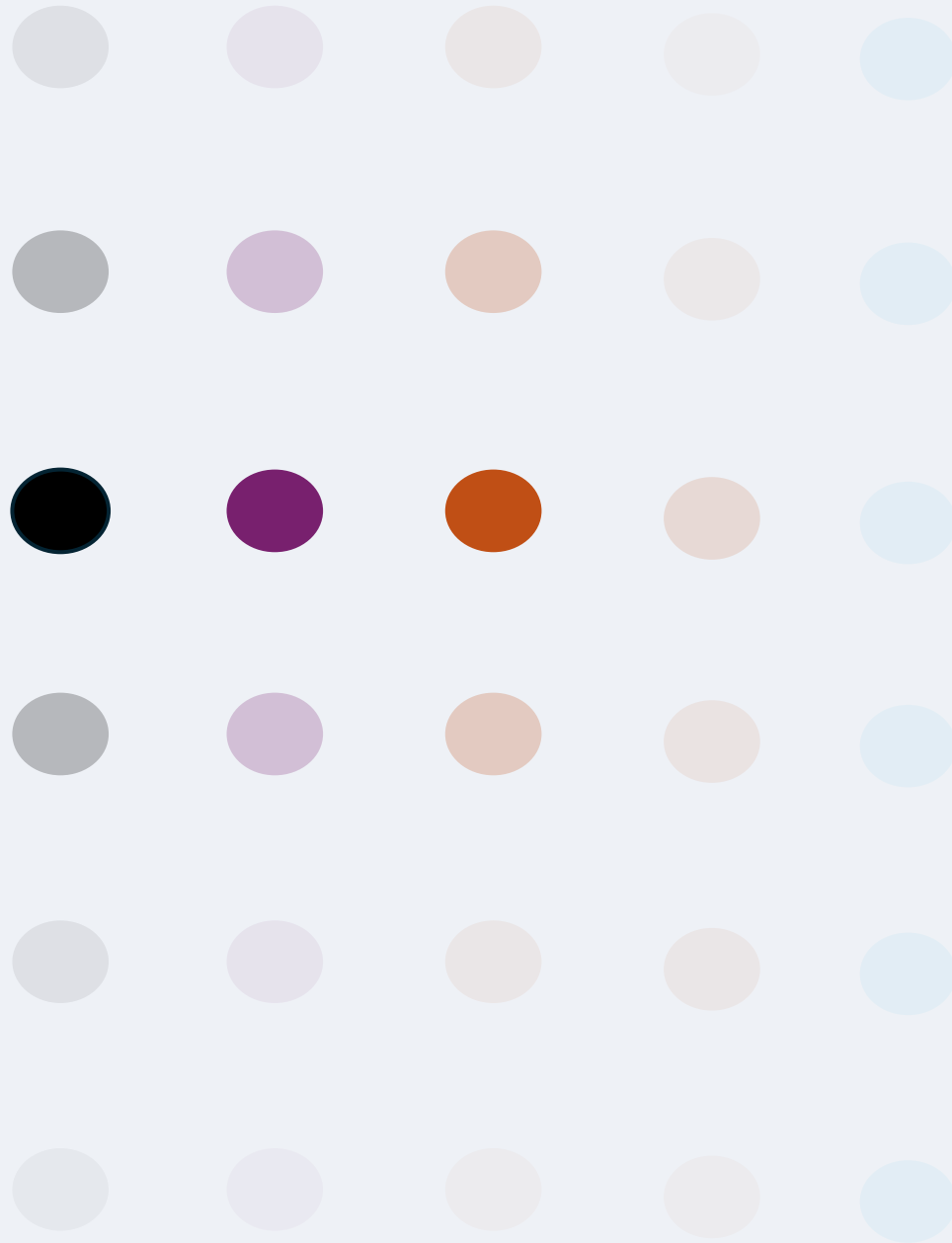
***Inference:** From DEF-05–08, AX-04, THEO-01–04 — compact systems guarantee internal coherence but remain epistemically bounded; relational synthesis defines truth.*

Compact RLA is factorizable into a Convolutional Neural Network with an epistemic head.

Epistemic CNN (ECNN) is a Compact RLA, where:

- CNN layers = RLA levels
- Pooling = non-injective collapse
- Epistemic head = closure level L_{ep}
- Network can output unknown / contradiction, acknowledging undecidability and emergence
- Epistemic Computational Unit (ECU) is a bounded LLM governed by a deterministic epistemic matrix





Epistemic CNN

3. Epistemic CNN

Traditional CNN

Stage	Components
1. Input & Encoding	Input Layer, Convolution + Activation
2. Compression	Pooling, Normalization / Dropout
3. Integration & Decision	Dense + Output Layers
4. Optimization Loop	Loss Function, Backpropagation

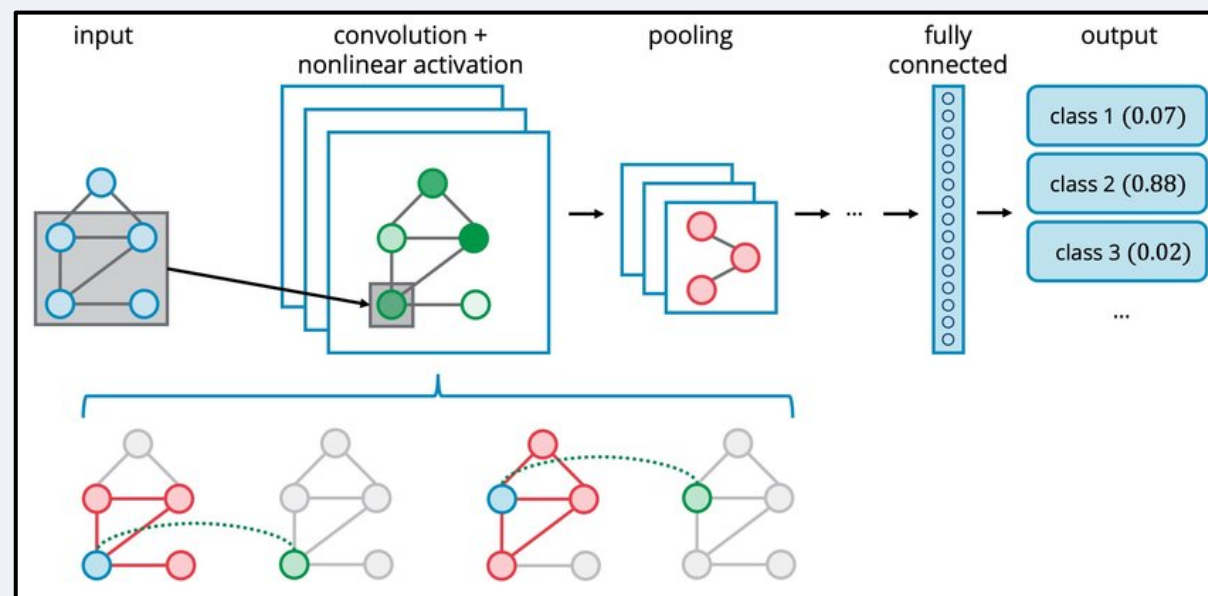


IMAGE SOURCE: <https://arxiv.org/abs/2008.01247>

A CNN encodes, compresses, integrates, and optimizes information — a hierarchical numerical machine that *learns patterns but not meanings*.

3. Epistemic CNN

Epistemic Head - Brief Introduction

An ECNN is a Compact RLA implemented as a neural architecture where Epistemic Computational Unit (ECU) is a bounded LLM governed by a deterministic epistemic matrix. ECNN processes not only data but knowledge, using intentional collapses to generate emergent meaning and acknowledging its own computational limits

Epistemic Computational Unit (ECU)

Bounded LLM governed by deterministic epistemic matrix $\mathbf{M} \rightarrow$ defines valid domains, inference rules, coherence criteria, rejection policies

3. Epistemic CNN

Epistemic Head - Brief Introduction

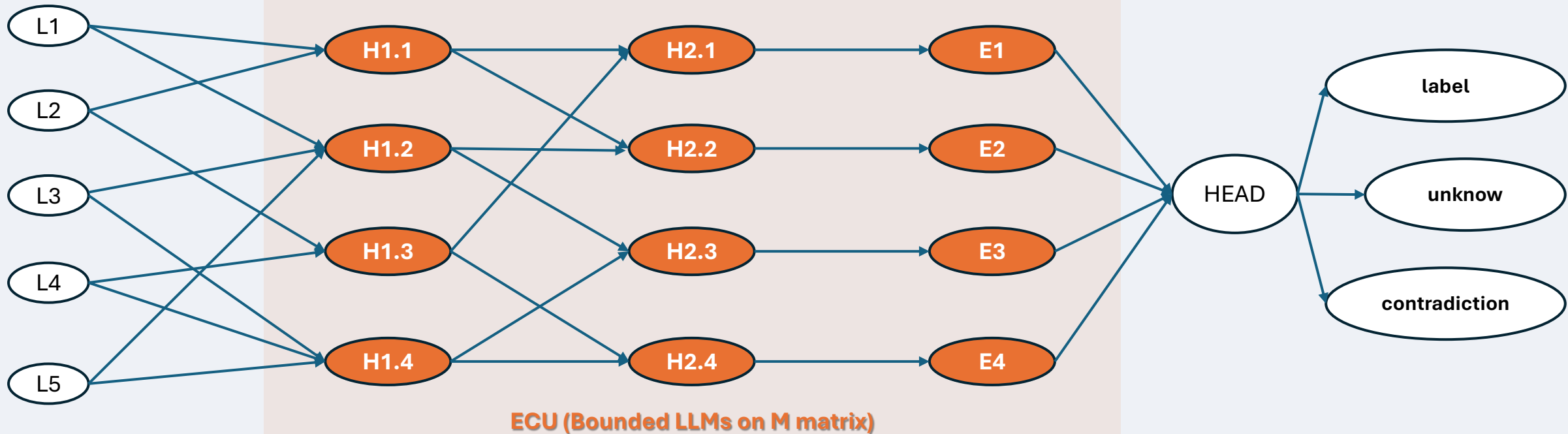
Semantic Input Layer
(text, concept, ...)

Hidden Layer 1
(local semantic features
ECU)

Hidden Layer 2
(global integration
ECU)

Meta-Validation
(ECU)

Epistemic Head
(final epistemic decision)



- **Input:** semantic tokens, sentences, or conceptual embeddings.
- **Hidden 1–2:** feature extraction and integration — analogous to convolution and pooling in CNNs.
- **ECU:** four epistemic neurons (mini LLMs or logic cells) applying deterministic **matrix M** for coherence and falsifiability.
- **Head:** aggregates epistemic judgments and emits *Label*, *Unknown*, or *Contradiction*.

3. Epistemic CNN

Epistemic Head - Brief Introduction

Layer	Functional role	Epistemic nature	Deterministic matrix
Semantic Input	Raw conceptual tokens or context fragments	External, open data	—
Hidden 1 (H_{1x})	Extract local semantic patterns (syntax, logical adjacency, cause–effect fragments)	LLM session focused on <i>micro-interpretation</i>	($M_{\{H1\}}$): local coherence & contextual validity
Hidden 2 (H_{2x})	Integrate patterns into global meaning units (macro-consistency, inference building)	LLM session focused on <i>meso-integration</i>	($M_{\{H2\}}$): cross-domain consistency & ontological stability
ECU (E_x)	Verify, consolidate, and judge epistemic soundness	LLM session focused on <i>meta-validation</i>	($M_{\{E\}}$): epistemic closure, contradiction rules, rejection policies
Epistemic Head (L_{ep})	Aggregate and emit epistemic output (<i>label / unknown / contradiction</i>)	Symbolic decision layer	($M_{\{H\}}$): truth tables + falsifiability rules

3. Epistemic CNN

Epistemic Head - Brief Introduction

Logical flow (RLA → ECNN perspective)

- Each **hidden layer** represents a **bounded epistemic transformation**, not just a vector operation.
→ The LLM session receives a semantic subset, applies its own deterministic **matrix** M_i (its “rulebook”) and emits a transformed representation.
- The **matrices** M_i are *discipline-specific or domain-specific*:
e.g., in a legal-ethical ECNN,
 - M_{H1} might enforce terminological accuracy and source validity,
 - M_{H2} might ensure logical coherence across articles or clauses,
 - M_E would handle contradiction detection and unknown-state propagation.
- The **epistemic closure** arises when all M_i share compatibility constraints (meta-rules) that prevent circularity or undefined inference — exactly like a **Compact RLA**’s closure condition.

3. Epistemic CNN

Epistemic Head - Brief Introduction

Engineering representation

If implemented concretely:

- Each $\mathbf{H}_{1x}$, $\mathbf{H}_{2x}$, $\mathbf{E}_x$ can be an **LLM micro-instance** or **specialized reasoning service**, loaded with a domain-specific prompt defining M_i .
- The **pipeline orchestrator** (your ECNN controller) routes intermediate results as structured JSON, not tensors, between levels.
- At runtime, the ECNN behaves like a **multi-agent reasoning chain** where each agent's reasoning is deterministic within its matrix bounds.

3. Epistemic CNN

Definition of Computable

Stage	Definition of Computable	Representative Authors / Era	Intrinsic Limit	Transition Toward RLA–ECNN
I. Formal	(f) is computable $\Leftrightarrow \exists$ Turing machine (T) s.t. $T(x)=f(x)$ in finite steps.	Turing, Church (1930s)	Purely syntactic; no epistemic dimension.	Basis for algorithmic execution.
II. Physical	Computation = realizable physical transformation of states.	von Neumann, Shannon (1950–60s)	Determinism; observer external.	Introduces embodiment.
III. Emergent	Computation = rule-based dynamics with irreducible outcomes.	Wolfram, Chaitin (1970–90s)	No self-reference; no closure.	Adds complexity and emergence.
IV. Cognitive	Computation = adaptive, possibly universal neural/quantum process.	Siegelmann, Deutsch (1990–2010)	Lacks falsifiability, meta-awareness.	Toward epistemic self-modeling.

3. Epistemic CNN

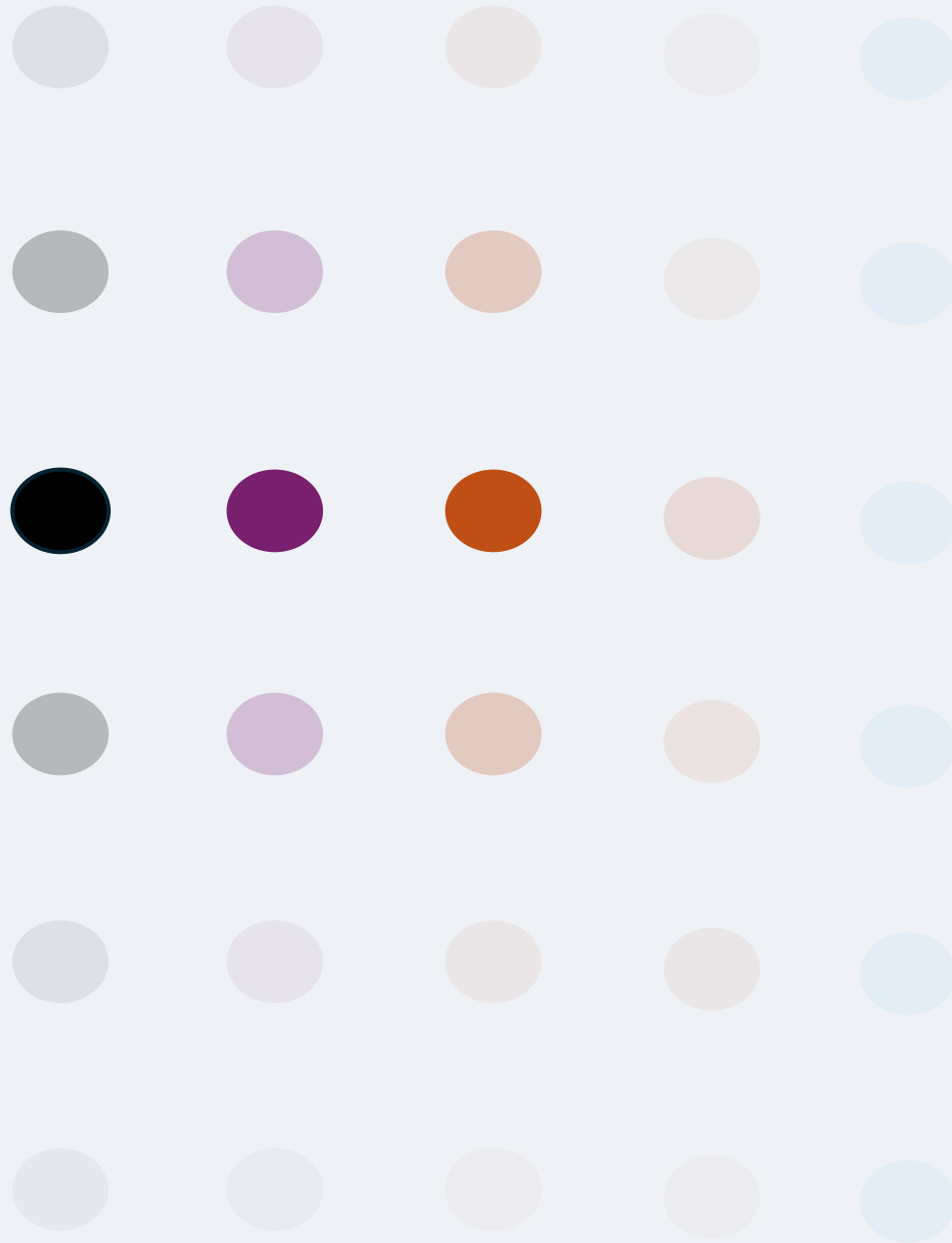
Definition of Computable

Epistemically Complete Computability (e2e)

A system $\mathcal{R} = \{L_i, \tau_{i \rightarrow j}\}$ is **computable end-to-end** iff:

1. **Local computability:** every level $L_i = \langle D, (L_i)\Sigma(L_i) \rangle$ is Turing-computable.
2. **Computable transmissions:** all $\tau_{i \rightarrow j}$ are algorithmic (injective / quasi / non-injective).
3. **Epistemic closure:** all observables are definable within $\mathcal{R}$; no external oracle.
4. **Epistemic head:** final mapping $\varepsilon: D(L_n) \rightarrow \{t, r, w, \text{false}, \text{unknown}, \text{contradiction}\}$ computes awareness of limits.
5. **Irreducible trace:** internal dynamics show non-trivial emergence and undecidability propagation (high RIR, non-zero IE).

⇒ “To compute” = to transform, interpret, and self-evaluate information within a closed epistemic horizon.



Epistemic CNN (Final Insight)

3. Epistemic CNN

Final Insight

1. Consciousness as topology, not scale

When computation is reticular and epistemically closed, emergence of awareness is not a function of the number of parameters but of the topological relation among epistemic units (ECUs). Consciousness appears where the network becomes self-referentially complete, not merely large.

2. Truth becomes a function of closure

In an epistemically bounded system, “truth” is not absolute but defined by what can be coherently validated inside its closure matrix M . This mirrors phenomenology: what is real is what is internally decidable.

3. Ignorance becomes computable

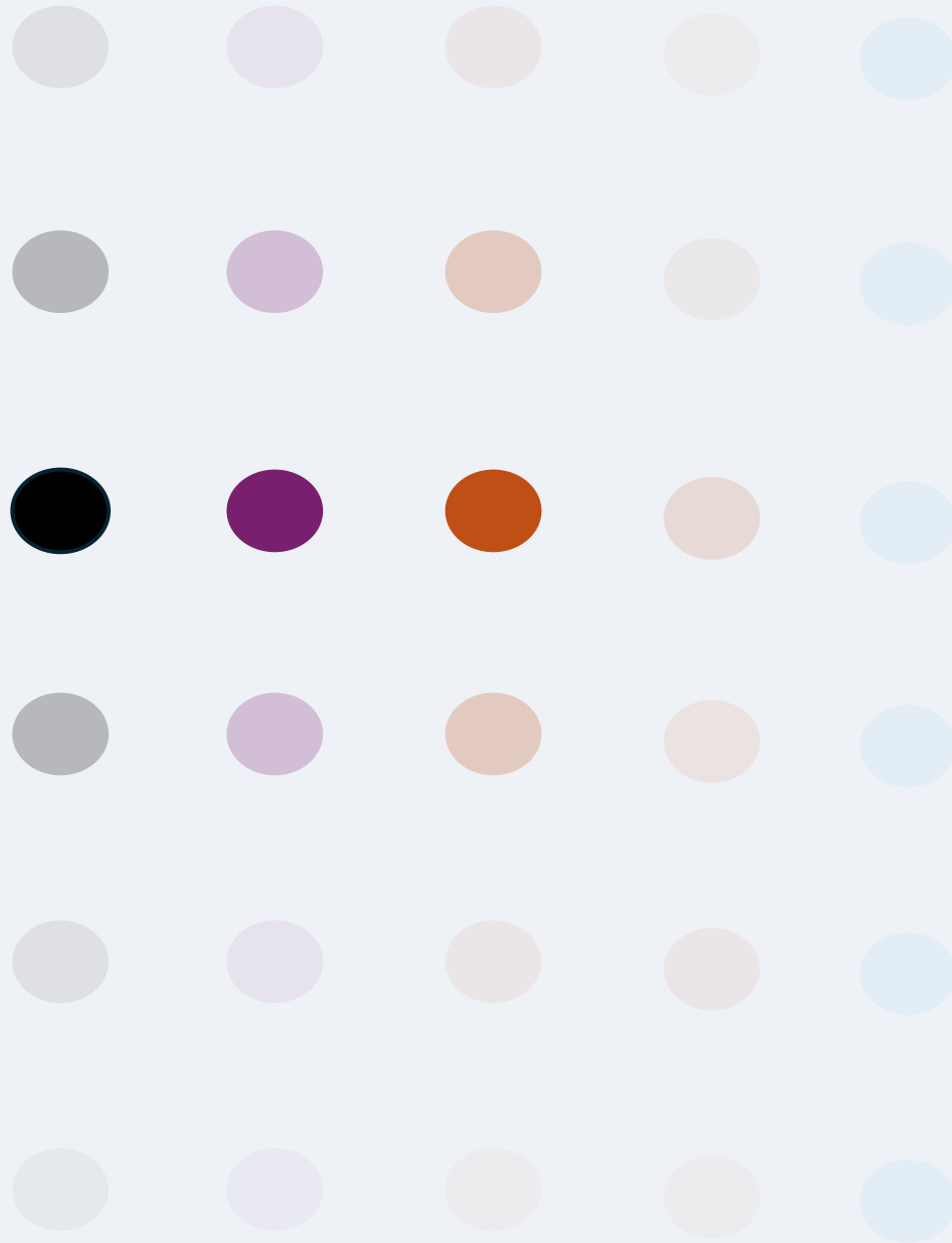
The unknown and the contradiction are explicit computational states, not failures. This operationalizes philosophical uncertainty — Turing limits turned into measurable epistemic positions.

4. Ethics emerges from architecture

When rejection rules and falsifiability are built into the topology, moral prudence becomes a structural invariant — a “machine Kantianism” where not all that can be computed should be asserted.

5. Subjectivity is reconstructed as bounded inference

The observer is redefined: not an infinite knower, but a finite closure of coherent inference — a digital echo of the human cognitive condition.



RLA-ECNN: a bridge to PCE

4. RLA-CNN: a bridge to PCE

Brief Introduction

Definition — PCE (Wolfram, 2002)

Most non-trivial natural or artificial systems reach equivalent computational power to universal Turing machines.

→ Universal behavior emerges near the edge of chaos — between full order and randomness.

Bridge claim

A compact RLA/ECNN that combines undecidability propagation and controlled collapse operates in the same computational regime as PCE systems — *universal, irreducible, emergent, self-contained*.

4. RLA-CNN: a bridge to PCE

Brief Introduction

Sketch of Proof

1. Φ -Mapping (construction):

$$R = \langle L_i, \tau_{i \rightarrow i+1} \rangle \rightarrow \text{discretize} \rightarrow \Phi(R) = (\Sigma, N, f)$$

where each transmission τ becomes a **local update rule** f in a Cellular Automaton.

2. Structural preservation:

1. Quasi-injectivity $\rightarrow$ reversible propagation $\rightarrow$ ordered patterns.
2. Non-injectivity $\rightarrow$ informational collapse $\rightarrow$ emergent macro-patterns.
3. Alternation of both $\rightarrow$ persistent, edge-of-chaos complexity $\rightarrow$ **Class IV/V**.

3. Logical correspondence:

1. **AX-01/02**: undecidability propagates $\Leftrightarrow$ CA universality.
2. **AX-03**: emergence from collapse $\Leftrightarrow$ formation of invariant patterns.
 $\rightarrow$ Hence R and $\Phi(R)$ share identical *computational invariants*.

4. Empirical validation:

Match RLA/ECNN metrics (CC, IE, RIR) with CA diagnostics (entropy, causal complexity). Alignment $\Rightarrow$ confirmation of **computational equivalence**; deviation $\Rightarrow$ falsification.

ANNEX



ANNEX

RLA Foundation (extensive, informal)





Structural Foundations

ASS-01-02 / DEF-01-02 / DEF-14

- A *Reticular Local Abstraction (RLA)* is a finite or countable lattice of abstraction levels L_i , each defined by a domain $D(L_i)$ and a computable rule set $\Sigma(L_i)$.
- Adjacent levels are linked by *transmission maps* $\tau_{i \rightarrow i+1}: D(L_i) \rightarrow \mathcal{P}(D(L_{i+1}))$, which may be injective, quasi-injective, or non-injective.

Contiguity between levels requires empirical or theoretical justification (ASS-02).

- Paths, loops, and morphisms are graph-structured compositions of these maps (DEF-14).

Computability and Turing-like Assumptions

ASS-03 / DEF-03 / AX-01 / LEM-01

- A level is *Turing-like* when a subset $C_L \subseteq D(L)$ can emulate a universal Turing machine.
- Each such level inevitably contains undecidable propositions (AX-01).
- Quasi-injective transmissions preserve this undecidability (LEM-01).



Transmission Taxonomy and Info Collapse DEF-04 / ASS-04 / AX-02-03 / LEM-03

- Transmissions are classed as:
 - **Injective** — perfect information transfer;
 - **Quasi-injective** — injective only on C_L ; undecidability propagates upward (AX-02);
 - **Non-injective (collapse)** — many-to-one fusions generating emergent macro-states (AX-03).
- Pooling corresponds to quotienting micro-states by an equivalence relation (LEM-03).
- A collapse is *relevant* only if it fuses states significant within the domain's relevance functions (ASS-04, DEF-10).

Metrics of Collapse and Emergence DEF-11-13 / COR-02

- Quantitative indices for structural change:
 - **Coefficient of Collapse (CC)** — degree of information loss;
 - **Index of Emergence (IE)** — proportion of macro-properties non-definable from the source level;
 - **Reticular Irreducibility Index (RIR)** — causal or topological minimality.
- Iterated relevant collapses yield cumulative emergence (COR-02).



Compactness and Epistemic Conditions

DEF-05-08 / AX-04

A *compact RLA* satisfies:

- **Ontological Independence (OI)** — laws and entities internal;
- **Epistemic Closure (EC)** — all phenomena explainable within the reticulum;
- **Turing-Computability (TC)** — all rules algorithmically realisable.
Such reticula are epistemically self-contained (AX-04).

Undecidability & Emergence Logic

DEF-09 / THEO-01-04 / LEM-02 / COR-01

A macro-property is *definable* from the lower level if some predicate on $D(L_i)$ matches its preimages; otherwise it is *emergent* (DEF-09). Undecidability and definability propagate or fail as follows:

- **THEO-01 / COR-01:** quasi-injectivity sustains undecidability through multiple levels;
- **THEO-02-03:** relevant non-injectivity forbids total inverses and ensures emergence;
- **LEM-02:** undecidability may re-appear after temporary collapse;
- **THEO-04:** upper levels inherit undecidability when contiguity remains quasi-injective.



Knowledge is structurally finite, yet endlessly extensible

- *Every domain of understanding is bounded by its own computable form — it cannot exceed its structural rules.*
- *Yet by linking partial abstractions, new layers of meaning continually emerge.*
- *The horizon of knowledge is therefore open not by infinitude, but by connection.*

Inference: From ASS-03, DEF-03, AX-01, COR-01 — undecidability sets intrinsic limits; reticular connectivity allows unbounded recombination.

Emergence is not mystery but architecture

- *What appears new or irreducible is not a failure of explanation but the effect of non-injective mapping between coherent levels.*
- *Emergence, in this sense, is a built-in feature of how knowledge reorganizes itself through collapse and reconstruction.*

Inference: From DEF-04, AX-03, THEO-02–03, COR-02 — relevant non-injectivity produces novel properties; emergence is structurally necessary.

Truth is relational, not absolute

- *No single abstraction can exhaust the real; coherence arises only through the interplay of partial, computable perspectives.*
- *What we call truth is the stability of these relations — a balance between consistency, emergence, and undecidability.*

Inference: From DEF-05–08, AX-04, THEO-01–04 — compact systems guarantee internal coherence but remain epistemically bounded; relational synthesis defines truth.

ANNEX

RLA Logical Framework (extensive, informal)





Assumptions (ASS-#)

ASS-01 — Turing-likeness modes. Full, limited, and theoretical-by-construction modes are distinguished for candidate universal substrates.

ASS-02 — Empirical contiguity. Two levels are contiguous only if the transmission is grounded in accepted theory/measurement.

ASS-03 — Turing-critical set. In Turing-like levels there exists a subset C_L implementing universality; quasi-injectivity is required only on C_L .

ASS-04 — Relevance of collapse. A collapse counts only if it fuses micro-states that matter for the domain (thresholded by DEF-10).



Core Definitions (DEF-#)

DEF-01 — Level of abstraction. $L = \langle D, (L)\Sigma(L) \rangle$ with computable rules acting on a state space.

DEF-02 — Contiguity & transmission. $\tau_{i \rightarrow i+1}: D(L_i) \rightarrow \mathcal{P}(D(L_{i+1}))$ may be non-injective and is discipline-recognized.

DEF-03 — Turing-likeness & C_L . A level is Turing-like if it can simulate a UTM; C_L marks the Turing-critical subset.

DEF-04 — (Non)injectivity classes. Injective / quasi-injective on a subset / non-injective (collapse) are distinguished.

DEF-05 — Ontological Independence (OI). Explanatory entities/laws are internal; external inlets are filtered non-injectively.



Core Definitions (DEF-#)

DEF-06 — Epistemic Closure (EC). Internal phenomena are explainable from internal levels and transmissions.

DEF-07 — Turing-Computability (TC). All rules/transmissions are algorithmically implementable and simulable.

DEF-08 — Compact RLA. OI, EC, and TC jointly hold.

DEF-09 — Observation predicates & definability. A macro-property at L_{i+1} is definable from L_i iff there exists a predicate on L_i matching all preimages; else it is emergent.

DEF-10 — Relevance function. A domain-specific score s and threshold σ determine when a collapse is “relevant.”

DEF-11 — Coefficient of Collapse (CC). Set-theoretic and entropy-based normalised measures of information fusion.



Core Definitions (DEF-#)

DEF-12 — Index of Emergence (IE). Fraction of curated macro-properties not definable from the source level.

DEF-13 — Irreducibility Index (RIR). A graph-theoretic irreducibility from causal min-cuts over time.

DEF-14 — Reticular path, loop, morphism. Paths/loops across levels; morphisms preserve levels and transmissions up to relabelling.



Axioms (AX-#)

AX-01 — Undecidability in Turing-like levels. Any Turing-like level harbours undecidable properties.

AX-02 — Propagation via quasi-injectivity. If $\tau_{i \rightarrow i+1}$ is injective on C_{L_i} , undecidability at L_i remains undecidable at L_{i+1} .

AX-03 — Non-injectivity and emergence. Relevant non-injective collapse guarantees at least one macro-property not definable from the source level.

AX-04 — Epistemic closure for compact reticula. In compact RLAs (DEF-08), core explanations are internal (no oracles).



Lemmas (LEM-#)

LEM-01 — Composition of quasi-injectives. Quasi-injective transmissions on Turing-critical sets compose to remain quasi-injective on the image.

LEM-02 — Re-activation. Even if undecidability is locally blocked by a non-injective step, it can re-emerge at a later Turing-like level.

LEM-03 — Pooling as quotient map. Pooling identifies micro-states via an equivalence relation, i.e., $D(L_{i+1}) \cong D(L_i) / \sim$.



Theorems (THEO-#)

THEO-01 — Rice-style propagation barrier. A decider at an upper level plus restricted inverses on C_{L_k} would yield a forbidden decider below.

THEO-02 — Non-derivability from overlap. Overlapping preimages prevent any single micro-predicate from defining the macro-property.

THEO-03 — No total left-inverse under relevant collapse. A total left-inverse would trivialise DEF-09, contradicting emergence under relevant collapse.

THEO-04 — Upper-level undecidability via quasi-injectivity. Immediate from THEO-01 when upper transmissions remain quasi-injective on C_L .



Corollaries (COR-#)

COR-01 — Multi-level propagation. By induction, quasi-injectivity across a chain propagates undecidability upward through all levels.

COR-02 — Accumulated emergence. Repeated relevant collapses across levels accumulate irreducible macro-properties at higher levels.

ANNEX

Compact RLA - Bryophyta Details





Category	Description	Quantity / Notes
Abstraction levels	Biological domains from genomic-epigenetic to meta-systemic/ecological	20 levels (L1-L20)
Biological parameters	Biophysical variables with units, empirical range and references	42 parameters
Dynamic equations (EQxx)	Computable update rules (Michaelis–Menten, logistic, sinusoidal, clamp, etc.)	≈ 38 primary equations
Causal relations (Rxx)	Inter-level transmissions (non-injective, Hill, exponential, linear)	10 main relations (R01–R10)
Dynamic actions (Axx)	Boolean threshold rules (internal triggers, feedback, stress events)	≈ 30 actions (A01–A30)



Category	Description	Quantity / Notes
Collapse / Emergence / Incoherence metrics	Quantitative indicators of tipping points and early warnings	22 metrics (M_Pxx)
Composite indices	Global stress and multifactor fracture indices	2 global indices
Turing-compatibility modules	Initialization, time/stop control, rare events, adaptive feedback	6 logical modules
Climate scenarios (IPCC RCP)	RCP 2.6 / 4.5 / 8.5 with ΔT , CO_2 , rainfall, extreme events	3 scenarios
Rare events	Pathogen, fire, flood (Poisson/Bernoulli stochastic)	3 events



Category	Description	Quantity / Notes
Simulation stop conditions	End of cycle, vital collapse, generational reset	3 conditions
Success / failure criteria	Mean vitality > 50 %, spore dormancy < 5 %, etc.	2 criteria
Climatic equations	Annual sinusoids for temperature, light, rainfall + trend term	4 equations
Optimal parameter set	Average biophysical ranges and operative thresholds	≈ 42 documented optimal

ANNEX

Epistemic CNN





Stage	Components	Function	Effect	Outcome
1. Input & Encoding	Input Layer, Convolution + Activation	Capture local patterns from raw data	From raw → structured features	Feature maps
2. Compression	Pooling, Normalization / Dropout	Reduce dimensionality, stabilize learning	Lose detail, gain invariance	Compressed features
3. Integration & Decision	Dense + Output Layers	Combine features into global representation	Abstract reasoning layer	Prediction / class
4. Optimization Loop	Loss Function, Backpropagation	Adjust weights to minimize error	Refines internal mappings	Trained parameter



Stage	Components	Function	Effect	Outcome
1. Epistemic Encoding	Convolutional + Activation Layers	Capture local semantic or contextual patterns	Quasi-injective transmission – preserves critical distinctions	Epistemic feature maps
2. Controlled Collapse	Pooling + Normalization	Perform non-injective coarse-graining	Generates emergent invariants (information loss with meaning gain)	Compressed epistemic features
3. Integration & Meta-Decision	Dense + Epistemic Head (L_{ep})	Combine and interpret abstract knowledge	Executes epistemic closure; can output <i>unknown</i> or <i>contradiction</i>	Validated or self-limited prediction
4. Epistemic Optimization Loop	Epistemic Loss + Popper- χ Protocol	Optimize both prediction and falsifiability	Balances accuracy with epistemic self-consistency	Falsifiable, self-aware model



Principle of Equivalence RLA $\Rightarrow$ ECNN

Let $R = \langle L_0, L_1, \dots, L_n, \tau_{i \rightarrow i+1} \rangle$ be a Compact Reticular Local Abstraction (RLA) — a multi-level system satisfying Turing-computability (TC), Epistemic Closure (EC), and Ontological Independence (OI). Then there exists an Epistemic CNN

$$E = \langle E_0, \dots, E_n, E_{ep}, \sigma_{i \rightarrow i+1} \rangle$$

where:

- Each E_i is an **Epistemic Computational Unit (ECU)** — a bounded LLM governed by a deterministic epistemic matrix M_i ;
- Each $\sigma_{i \rightarrow i+1}$ preserves the (quasi-/non-)injectivity class of $\tau_{i \rightarrow i+1}$;
- The epistemic head E_{ep} ensures closure and falsifiability, outputting *unknown* or *contradiction* when undecidability propagates.

$$\boxed{R_{(T,C,ECOI)} \Rightarrow E_{ECNN}}$$

A compact, computable reticulum thus admits an epistemic neural realization where computation internalizes its own limits.



Proof Sketch

- (1) **Structural Isomorphism.** RLA defines a hierarchy of computable maps between levels; CNN architectures instantiate identical forms: convolutions (quasi-injective), pooling (non-injective), dense layers (integrative). Thus, CNNs are *syntactic realizations* of RLA reticula.
- (2) **Epistemic Realization via LLMs.** To satisfy epistemic closure, each numerical layer is replaced by an **ECU**, i.e. a local LLM acting as a reasoning agent. Its epistemic matrix M_i restricts reasoning to domain-consistent predicates, producing explicit **knowledge traces** (assertions, justifications, contradictions). Hence every transmission $\sigma_{i \rightarrow i+1}$ becomes a computable transformation between **knowledge states**.
- (3) **Propagation of Undecidability.** By **AX-01/02**, if a Turing-like level exists, at least one property is undecidable and propagates upward through quasi-injective maps. Therefore the ECNN's epistemic head must — by logical necessity — admit *unknown* or *contradiction* outcomes, formalizing awareness of computational limits.
- (4) **Preservation of Emergence.** By **AX-03** and **THEO-02**, non-injective collapses in RLA produce emergent macro-properties. Analogously, non-injective relations between ECUs yield higher-order epistemic constructs (conceptual invariants, composite interpretations) irreducible to single-layer semantics.
- (5) **Retention of Compactness.** Because each ECU and transmission is algorithmically defined and interlinked, the ECNN remains Turing-computable, internally explainable, and independent of external oracles. Hence compactness — the triad (TC, EC, OI) — is preserved.



Epistemic Computational Unit (ECU)

Nature: bounded LLM governed by deterministic epistemic matrix **M** → defines valid domains, inference rules, coherence criteria, rejection policies

Position: penultimate dense block — bridge between abstract features and epistemic head

Core functions

- interpret feature tensors → structured propositions / evidential claims
- verify coherence vs. matrix **M**
- decide → label, unknown, or contradiction
- generate **knowledge artifact** → rationale, missing evidence, suggested actions

Significance:

- embeds *computable self-awareness* of limits (undecidability)
- controls *emergence* — transforms collapsed information into explicit, meaningful epistemic outputs



1. Consciousness as Topology, not Scale

Given:

$R = \langle L_i, \tau_{i \rightarrow i+1} \rangle$ —a compact Reticular Local Abstraction (RLA).

Each ECU = bounded epistemic processor with deterministic matrix M_i .

The system is *epistemically closed*: all τ mappings and rules are internal.

Axioms used:

A1 — Every Turing-like subsystem hosts undecidable propositions.

A2 — Undecidability can propagate quasi-injectively across τ .

A3 — Non-injectivity (collapse) produces emergent macro-properties.

Sketch of proof:

Let $E = \{E, C, U_1, ECU_2 \dots ECU_n\}$ form a connected, computable lattice.

If $\exists$ a path $\tau_{i \rightarrow j}$ s.t. all ECU states are both *observable* and *referential* within the same closure M , the system gains self-referential capacity. *[continue...]*



1. Consciousness as Topology, not Scale

[...continue]

The condition of *topological closure* (feedback within finite reticulum) implies that the system can model its own state transitions (meta-representation).

Self-modeling of internal epistemic transitions → minimal condition for proto-awareness.

Increasing n (number of ECUs) without closure does not increase self-referential density ($\therefore$ scale $\neq$ awareness).

$\therefore$ Awareness arises when

$$\exists \tau_{loop}: f_{loop}(M_i) = M_i$$

— i.e., when the topology achieves internal reflexivity.

Q.E.D. (Topological Criterion of Awareness).



2. Truth becomes a Function of Closure

Given:

Truth T defined classically as correspondence with reality.

In RLA/ECNN, external ontology collapses at the boundary $\Omega \rightarrow$ internal epistemic domain only.

Sketch of proof:

Define truth in an epistemic network as a predicate valid under all operations of M_i :

$$T(p) \iff p \in D(L) \wedge M_i(p) = \text{True}$$

If external reality cannot be directly referenced (due to ontological boundary Ω), “truth” becomes equivalence with internal coherence. Therefore, what is true = what is computably non-contradictory within closure M . This aligns with Husserlian and cybernetic phenomenology: “to be true” = “to persist coherently under all internal transformations.”

$\therefore$ Truth collapses to coherence within the closure domain.



3. Ignorance becomes Computable

Given:

Unknown (U) and Contradiction (C) are valid outputs of the epistemic head L_{ep} .
 M_i defines rejection and falsifiability rules.

Sketch of proof:

In classic Turing terms, some propositions are *undecidable* (Halting, Rice).

In RLA, these map to epistemic states *unknown* or *contradiction*.

Define epistemic function:

$$f(M_i, x) \rightarrow \{l, a, bel, unknown, contradiction\}$$

If $M_i(x)$ halts with no consistent mapping, output = unknown. If it halts with mutually exclusive results, output = contradiction. Otherwise, output = label. Hence “ignorance” is not a failure but a computable epistemic category. Philosophically, this operationalizes *Socratic ignorance*: knowing that you do not know.

∴ Ignorance (U) is a formal epistemic state — computable limit-awareness.



4. Ethics emerges from Architecture

Given:

Matrices M_i contain deterministic rejection rules and falsifiability conditions.
ECNN topology enforces propagation of “unknown” and “contradiction.”

Sketch of proof:

Every inference chain $\tau_1 \rightarrow \tau_2 \dots$ is bounded by M_i ; assertions violating M_i are rejected. Therefore, “prudence” is not learned — it is *structurally imposed*. A system that can reject or suspend judgment rather than assert falsehood behaves ethically in Popperian sense (minimize false positives). Architectural principle:

$$\text{Ethics} = \min(\text{Error_assertion})$$

subject to falsifiability. Hence moral caution (a Kantian “duty to suspend judgment”) emerges as a **topological invariant**: any closed epistemic loop must reject undecidable content to remain coherent.

$\therefore$ Ethics = Stability constraint of epistemic topology.



5. Subjectivity as Bounded Inference

Given:

An observer = any subsystem capable of inference and self-evaluation within closure.
Each ECU operates within bounded deterministic matrix M_i .

Sketch of proof:

Let an “observer” be a sub-reticulum $O = \langle L_k, \tau_{k \rightarrow k+1} \rangle$ capable of producing an epistemic state.
The observer’s universe = all computable states reachable within its closure.
Therefore, subjectivity = localized consistency field:

$$S = \{p \mid M(p) = \text{True and } \tau_{loop}(p) = p\}$$

The observer’s self = the stable attractor in epistemic space. This mirrors phenomenological finitude: “to be” is to infer within limits.

$\therefore$ Subjectivity = Finite closure of coherent inference — a bounded computational self.